
A PIPELINE FOR MULTIPLE ORANGE DETECTION AND TRACKING WITH 3-D FRUIT RELOCALIZATION AND NEURAL-NET BASED YIELD REGRESSION IN COMMERCIAL CITRUS ORCHARDS

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ABSTRACT

Traditionally, sweet orange crop forecasting has involved manually counting fruits from numerous trees, which is a labor-intensive process. Automatic systems for fruit counting based on proximal imaging, computer vision, and machine learning, have been considered a promising alternative or complement to manual counting. These systems require data association components that prevent multiple counting of the same fruit observed in different images. However, there is a lack of work evaluating the accuracy of multiple fruit counting, especially considering (i) occluded and re-entering green fruits on leafy trees, and (ii) counting ground-truth data measured in the crop field. Here, we propose a non-invasive alternative that utilizes fruit counting from videos, implemented as a pipeline. Firstly, we employ convolutional neural networks for the detection of visible fruits. Inter-frame association techniques are then applied to track the fruits across frames. To handle occluded and re-appeared fruit, we introduce a relocalization component that employs 3-D estimation of fruit locations. Finally, a neural network regressor is utilized to estimate the total number of fruit, integrating image-based fruit counting with other tree data such as crop variety and tree size. The results demonstrate that the performance of our approach is closely tied to the quality of the field-collected videos. By ensuring that at least 30% of the fruit is accurately detected, tracked, and counted, our yield regressor achieves an impressive coefficient of determination of 0.85. To the best of our knowledge, this study represents one of the few endeavors in fruit estimation that incorporates manual fruit counting as a reference point for evaluation. We also introduce annotated datasets for multiple orange tracking (MORANGET) and orange detection (ORANDET), which are publicly available and aim to foster the development of novel methods for image-based fruit counting.

Keywords Orange fruit detection · Orange fruit counting · Multiple-object tracking · Crop yield estimation · Deep learning · Digital agriculture

1 Introduction

Accurate fruit yield estimation is important for making informed decisions about harvesting, storage, and marketing, but such estimation can be a challenging task. Consider the major citrus belt in Brazil, which covers areas in two states (São Paulo and Minas Gerais) and sums 461,921 hectares, 86% reserved to sweet orange production, according to the Fund for Citrus Protection, Fundecitrus [2022]. The most used annual orange crop forecast process, performed by Fundecitrus, is a laborious operation that involves manual fruit stripping (the advanced harvest of all fruit in the tree), in

samples of around 1,500 trees spread across this citrus belt. Such sampling and counting procedures are a usual way to obtain pre-harvest fruit yield data [Wulfsohn et al., 2012, He et al., 2022a].

Computer vision-based methods have been considered as a prominent alternative for automatic, non-invasive fruit counting, becoming parts of larger yield estimation systems. Earlier works employed *feature engineering* [Gongal et al., 2015] to create fruit detectors, exploring low-level features in images derived from color and texture. An example for citrus was presented by Maldonado and Barbosa [2016], that employed image processing techniques and support vector machines to detect green oranges in field images. With the popularization of deep learning, convolutional neural networks (CNNs) were adopted for automatic feature learning (*representation learning*) [LeCun et al., 2015]. The work by Sa et al. [2016] is a turning point in the adoption of CNNs in fruits detection, and this passage from feature engineering to representation learning is reviewed by Koirala et al. [2019b].

Despite significant progress in automated fruit detection using deep learning techniques, the *counting* stage of the processing pipeline remains a challenging area that requires further development. Single images cannot provide reliable estimates of fruit counting because of occluded fruit: a single standpoint keeps a significant part of the fruit out of sight [Wang et al., 2019]. A natural alternative is to employ multiple images (or, equivalently, video sequences), to get different views able to reach most of the targets and perform counting of the detected fruits. However, while single-view fruit counting presents underestimation issues, a major limitation of multiple-view fruit counting is overestimation of yield due to overlapping images, which must be properly managed to ensure accurate counting [Villacrés et al., 2023]. An approach being explored recently is to scan the orchards using cameras and to assign a unique identifier to each fruit along the image sequence [de Jong et al., 2022, Villacrés et al., 2023]. Such approach, *Multi Object Tracking (MOT)*, involves detecting multiple objects in a sequence of images and linking them over time to identify individual objects [Luiten et al., 2021]. This is a challenging task that requires precise detection, localization, and association of a same object in different images. Essentially, the goal is to identify each object’s location, movement, and identity throughout the entire sequence, avoiding multiple counting of the same object (a fruit). Fruit detection and tracking can also be considered a component of *semantic scene understanding* systems¹ for agricultural settings, allowing artificial agents to extract semantic information about the objects in an orchard and interact usefully with its environment. Examples of such interactions are crop monitoring, spraying and harvesting [Duckett et al., 2018].

Our objective is to contribute to crop forecast by providing estimates of the actual number of fruits on a tree. We start with the challenging task of *counting visible oranges* on individual trees under real citrus farm conditions using video inputs. Yield prediction is performed in earlier maturation stages of sweet orange, when the fruit is green (non-mature), lacking the color contrast against green leaves observed in mature fruit [Gong et al., 2013]. The imaging is performed on ground level, between trees rows, employing smartphone cameras. In such a setting, there is an issue: tall trees (up to 5 meters height) cannot be fully imaged in a single frame. In other words, considering the available inter-row space, the trees’ height and the absence of wide-angle optics, it is not possible to keep the entire tree in the camera’s field of view (FOV). Even in the case of wide-angle optics availability, a single view of a tree cannot provide a full assessment of the entire plant fruit set because of occlusions. Therefore, we need of a multiple-view counting system to solve this issue. After fruit counting, we proceed to the *yield estimation* for each tree. This value differs from the counted fruit by any computer vision system, since there are oranges located in the inner part of the plant canopy, not visible in the images collected in the field even considering multiple views. We implement a neural net-based regressor to estimate the actual number of fruit from the fruit counting, integrating other data as plant height, variety, and age. Although we use a different set of methods, our work is similar to the one by Häni et al. [2020] in that they also propose a pipeline comprising fruit detection, 3-D projection, fruit counting and yield estimation in apple orchards. However, our study faces a more challenging scenario considering occlusions and fruit tracking.

The main contributions of this work are:

- a simple but effective annotation methodology for multiple object tracking in citrus, exploiting camera pose data by structure from motion to estimate the fruits’ 3-D location and automate the generation of 2-D tracks ground-truth;
- the introduction of MORANGET [Santos et al., 2023], a public MOT dataset for citrus composed by images sequences (frames) recorded on citrus farms plus ground-truth tracking data in the MOT16 format [Milan et al., 2016], whose annotation effort was performed by our team;
- a multiple fruit detection and tracking method based on convolutional networks for detection, on the Hungarian algorithm [Kuhn, 1955] for inter-frame association, and a relocalization algorithm to treat occlusions and fruits exiting/entering the camera’s field of view;
- an evaluation of the proposed method employing the *multiple object tracking accuracy* (MOTA) [Bernardin and Stiefelwagen, 2008] and the *higher order tracking accuracy* (HOTA) [Luiten et al., 2021], and

¹Or *SpatialAI* systems, as proposed by Davison [2018].

- proposition of a regressor based on artificial neural networks to estimate the number of fruit in a tree given the number of fruit counted using MOT, validated using a set of 1,139 trees with real yield manually collected.

A useful property of the proposed tracking method is the fruit’s relative position in 3-D is also estimated, allowing assessment of the spatial distribution of fruits in the tree.

This work is organized as follows. In Section 2 we provide a comprehensive overview of related works in the field of fruit counting and yield estimation. We discuss the various methods employed for fruit detection and inter-frame fruit association. Section 3 presents our proposed methodology, which encompasses the MORANGET dataset, a novel annotation technique for MOT ground-truth, the methods employed for orange tracking, and tree yield estimation. The results of our experiments are presented in Section 4, followed by a detailed discussion of the findings in Section 5. Finally, in Section 6, we present our concluding remarks, summarizing the main contributions of this work and outlining potential paths for future research. For a visual overview, a video showing our fruit counting results is available².

2 Related work

After the raise of CNN-based methods for fruit detection, initiated by Sa et al. [2016], researchers have explored the problem of fruit counting in multiple images, considering important challenges as fruit recounting and occlusions [Wang et al., 2019, de Jong et al., 2022, Villacr s et al., 2023]. Most of these works add a *data association* procedure to link detected oranges in different images, avoiding multiple counting, and commonly employing a tracking algorithm. Some researchers extended their fruit counting systems to yield estimation [Liu et al., 2019, H ni et al., 2020, Koirala et al., 2021], generally using (linear) regression.

Liu et al. [2019] proposed a fruit counting system tested on data from mangoes orchards. A dataset of 1, 500 images, presenting a resolution of 500×500 pixels, was employed to train a Faster R-CNN detector [Ren et al., 2017]. The mangoes’ tracking was performed by a method employing Kanade-Lucas-Tomasi algorithm (*optical flow* or KLT), Kalman filters, and the Hungarian algorithm [Kuhn, 1955]. To deal with long-time occlusions and reappearing fruits, they employed a relocalization procedure based on the fruit’s estimated 3-D position and reprojection on the current frame. They used *structure from motion* (SfM) [Dellaert, 2021], using the mangoes as *3-D landmarks* and their tracked positions in images as *measurements*. In SfM, the three-dimensional positions of landmarks and the camera are estimated from the measurements (2-D positions in each frame) using non-linear least squares [Hartley and Zisserman, 2004, Dellaert, 2021]. Liu *et al.* employed COLMAP [Schonberger and Frahm, 2016] as SfM implementation, getting estimations for the mangoes locations in 3-D and the camera position in space at each frame. To avoid multiple counting, mangoes/landmarks and Faster R-CNN detected bounding boxes are reassociated using the Hungarian algorithm and a specially designed association cost function. Liu *et al.* achieved a coefficient of determination (R^2) of 0.78 in a linear regression model relating fruit count to yield after considering views from two different vantage points of two rows facing opposite sides of the trees.

Concurrently, Wang et al. [2019] proposed another image-based counting system for mangoes. Using 10 fps video sequences scaled to 1024×1024 pixels frames, they employed their MangoYOLO model, a 33-layers adaption of YOLOv3 [Koirala et al., 2019a], to detect fruits. Kalman filter was employed on tracking, estimating the fruit positions in the next frame. The Hungarian algorithm was used to assign predicted fruit positions to detections in the current frame, using the fruits’ centroids distances on the cost function. Wang et al. adopted a threshold for lost trackers: after 15 frames without associations to MangoYOLO detections, the Kalman-based tracker is ended – if the fruit re-appear, it is considered a new fruit, incrementing counting. In an experiment involving a row containing 21 trees, the authors reported a root-mean-square error (RMSE) of 18 fruits/tree. However, reported results indicate a canceling occurring in the mean error between overestimation by repeated counting and underestimation by missed fruits.

Gan et al. [2020] proposed a novel approach where a vehicle carrying nozzles applied a water mist to citrus trees. The mist induced temperature contrast between fruits and leaves and a thermal camera, also embedded in the vehicle, recorded video streams of thermal frames. Careful experimentation was conducted to find optimal parameters for the amount of water and kind of nozzle, and the impact of ambient temperature and humidity considering the goal of maximizing temperature contrast. A Faster R-CNN model was trained for orange detection in the thermal images, reaching 87.2% average precision (AP). KLT tracking was employed to avoid multiple counting, but the data association used by the authors for multiple fruit tracking is unclear. In a row containing 25 trees, their method was able to count 96% of the 747 manually counted fruits in the field. However, the thermal camera’s FOV was unable to cover the entire canopy, so the manually collected ground-truth was restricted to the view area, marked using nylon strings.

Gao et al. [2022] have explored the modern fruiting-wall growing system for apples, where the apple tree canopy is thin, making the fruits and trunks visible. Fruit detection and tree trunk detection were performed using the YOLOv4-tiny

²<https://youtu.be/h0q42KmskLQ>

architecture [Bochkovskiy et al., 2020], a fast neural network aimed for real-time applications. They assume a linear movement of the camera (embedded in a vehicle), so their tracking is adapted to deal with horizontal movements only. The CSR-DCF algorithm [Lukezic et al., 2017], a single target tracker, was employed on trunk tracking, and the estimated horizontal movement guided the apples tracking. Inter-frame fruit assignment is guided by minimum Euclidean distance between fruits’ centroids, but the employed assignment algorithm is not completely clear. The counting system, evaluated using annotated videos (not the true yield or the manual counting on the field), reached $R^2 = 0.98$. However, the reported results show detection errors in a single frame (mainly false negatives) produce immediate tracking errors.

The APPLE MOTS dataset was introduced by de Jong et al. [2022]. Two UAVs and a wearable device, where a camera is embedded in a helmet, were employed to get video sequences in orchards presenting three different apple varieties. The CVAT annotation tool was employed to produce almost 86,000 manually annotated masks associated to 2,304 unique apple instances across 1,673 frames. The authors further evaluated two different MOT algorithms, TrackR-CNN [Voigtlaender et al., 2019] and PointTrack [Xu et al., 2020]. The best results were produced by PointTrack, that reached 52.9 in the multi-object tracking and segmentation accuracy (MOTSA) metric, a variation of the MOT metric [Milan et al., 2016] for segmented objects³. The authors call attention to the fact that multiple fruit tracking involves *homogeneous objects*, i.e., objects very similar to each other. This differs from the people tracking, a popular task in computer vision because of research on surveillance and autonomous vehicles. De Jong et al. consider if this homogeneity in object appearance could impact methods developed originally for pedestrian tracking in different ways.

Another work employing a MOT formulation was presented by He et al. [2022b], again following a tracking by detection framework: a YOLOv3 network [Redmon and Farhadi, 2018] was employed to find camellia fruit and apples, and Kalman filters used on fruit tracking. The Mahalanobis distance and the Kalman-based predictions are employed to define eligible associations between tracks and new detections in the current frame. Then a similarity function based on appearance is employed on association by a nearest-neighbor algorithm. Such a combination of spatial filtering and appearance similarity was named *Cascade-SORT* by the authors. He et al. [2022b], have reached MOTA values beyond 0.70 for most of their tests. However, their results are composed of just four short video sequences (one for camellia and three for apples).

Zhang et al. [2022] proposed a multiple object tracking method for citrus in-field, composed of a detection component, OrangeYOLO, and a tracking component, OrangeSORT, which are modified versions of the YOLOv3 network [Redmon and Farhadi, 2018] and the SORT algorithm [Bewley et al., 2016], respectively. A wide-angle action camera⁴ was adapted to a field rover, that performed a linear movement across orchards’ rows at 2 m/s. Zhang *et al.* carefully selected three scales from YOLOv3’s backbone (Darknet53) after analyzing the receptive fields and the oranges’ sizes, getting significant improvements in detection. OrangeYOLO also presents a channel-spatial attention mechanism, but the ablation experiment results showed a marginal gain in detection by this extension to the architecture. OrangeSORT computes the average motion displacement observed in the tracked fruits to update the state of lost trackers (Kalman filters), i.e., trackers not associated to any OrangeYOLO’s detection at the current frame. In the six videos collected in-field for testing, counting error varied from 0.61% (the best case) to 24.29% (worst case).

Villacrés et al. [2023] evaluated five different MOT algorithms for tracking apples in-field: (i) multiple Kalman filters combined with the Hungarian algorithm; (ii) kernelized correlation filter; (iii) multiple hypothesis tracking (MHT) [Kim et al., 2015]; (iv) SORT [Bewley et al., 2016] and (v) DeepSORT [Wojke et al., 2018]. They trained the employed apple detectors, Faster R-CNN and YOLOv5, using public datasets as MinneApple [Hani et al., 2020] and Fuji-SfM [Gené-Mola et al., 2020]. Nine videos recorded in-field using smartphones⁵ were employed on the apples tracking evaluation. Annotations in the MOT format [Milan et al., 2016] for the videos were produced using the open-source tool CVAT⁶. The authors conducted an interesting sensitivity analysis, employing the ground-truth bounding boxes as input for the trackers, but degrading the detection probability from 100% (all boxes in the ground-truth) to 20%. MHT and DeepSORT showed the best tracking performances, even at 60% detection rate. When considering detections from the neural nets, YOLOv5 produced the best detections for tracking, while DeepSORT produced the best results: 20.1% error on average for fruit counting.

What is the proper way to evaluate MOT trackers? According to Luiten et al. [2021], evaluation metrics serve two primary purposes. Firstly, they allow for straightforward comparison between various methods to determine which ones perform better than others. To achieve this objective, it is recommended to have a single metric that can be used to rank and compare these methods. Secondly, evaluation metrics are essential for analyzing and comprehending the various types of errors that algorithms make, identifying where they are likely to fall short when applied. The HOTA metric

³In the segmentation case, *masks* are provided as object segmentation ground-truth, not just the rectangular bounding boxes.

⁴A DJI Osmo Action camera, presenting 145° FOV.

⁵Smartphones iPhone 6S Plus and iPhone 13, Apple Inc.

⁶CVAT – <http://cvat.ai>.

[Luiten et al., 2021] was proposed with these two objectives in mind for the complex task of multiple object tracking. As stated by its proponents:

HOTA measures how well the trajectories of matching detections align, and averages this over all matching detections, while also penalizing detections that don't match.

Few works have employed proper MOT metrics for multiple fruit tracking [de Jong et al., 2022, Villacrés et al., 2023], most of them employing the *multiple object tracking accuracy* (MOTA) metric proposed by Bernardin and Stiefelhagen [2008]. However, MOTA has been criticized because it tends to prioritize detection over association [Luiten et al., 2021], while HOTA offers a balanced combination of detection and association scores.

Lost detections, caused by occlusions and the harsh light conditions in orchards, lead to fruit track breaks and yield overestimation, as noted by He et al. [2022b] and observed in results by Gao et al. [2022]. We argue that a *long-term process*, as relocalization, is needed for fruit counting, providing robustness to *short-term* detection association in neighbor video frames. The same was speculated by Villacrés et al. [2023] while Liu et al. [2019] developed their landmark-based re-association method to tackle this issue. The present work is closely related to [Liu et al., 2019]: the 3-D information is employed to, using geometrical restrictions, properly re-assigned lost fruits, and both works use SfM to get 3-D information. However, Liu et al. [2019] use the fruits as landmarks, an interesting approach to reduce SfM complexity, but it raises questions about SfM performance when only a few fruits (or none) are seen by the camera. In such case, the SfM framework could not have landmarks to proper camera motion estimation. In our method, SfM is performed to get the *camera relative motion* (ego-motion), and fruit localization in 3-D is performed as an independent process. The tracking method presented in our work, however, is more similar to SORT-based methods, in the sense the assignments are performed online if ego-motion data is available, differently from the offline SfM processing in Liu et al. [2019]. Our approach is complementary to the method presented by He et al. [2022b], which assumed unknown camera position (no ego-motion) and employed Kalman filter-based tracking to predict the trajectories of occluded or misdetections. Conversely, we explore relocalization for long-term tracking, based not on appearance, but on 3-D localization estimated using ego-motion data.

The methods for yield estimation vary depending on the type of data collected or the crop. Wulfsohn et al. [2012] sample trees from an orchard and select branches and segments of branches using a multi-stage systematic sampling approach. They then apply statistical methods to propose a yield estimator. On the other hand, Häni et al. [2020] used machine learning and tracking to provide a direct estimation of fruits. In their case, the structure of the plants makes the fruit visible on the surface of the canopy, allowing direct counting. Direct estimation of fruits based on images was also performed by Maldonado and Barbosa [2016]. In our problem, direct estimation is not viable because we identified a large gap between the visible fruits and the total number of fruits obtained through manual harvesting. This gap may result from the fact that some fruit is located inside the canopy. To address this issue, we propose a neural network regressor for the yield that considers the number of identified fruits and other relevant variables, as detailed in Section 3.4. A similar approach was used in estimating mango fruit load, where Koirala et al. [2019a] applied a correction factor to estimate the fruit load per orchard. However, unlike the regressor we are proposing, this correction factor is calculated per orchard based on the average ratio of the number of fruits identified in images to the manual harvest count per tree in the field.

3 Materials and methods

Our pipeline takes video files that record both faces of a tree, one for each side of the row, in the field. It then generates an estimation of the total yield for that tree, measured in the number of fruits. Figure 1 shows a pipeline overview. Videos are recorded by field staff (Section 3.1.2), and ID numeric plates are displayed in the first frames to identify each tree. The first step of the pipeline is to identify the plate and extract the tree ID t . The entire sequence of video frames is extracted from the video file, and a frame sampling procedure (Section 3.1.3) selects a subset. As the recording is performed by handheld cameras (smartphones), and the human camera operator slowly films the tree, several frames can be very similar because of too slow camera motion: the frame selection employs the well-known framework of feature detection and matching [Muja and Lowe, 2014] to select frames presenting distinguished changes, but avoiding large movement that could prejudice ego-motion estimation and fruit tracking. The selected frames feed a structure from motion procedure that estimates the camera parameters, including position and orientation, for each frame (Section 3.1.4). The selected frames are also inputted to a fruit detection neural network, producing a set of rectangular bounding boxes that mark the observed oranges in each frame (Section 3.2). Our fruit tracking procedure (Section 3.3) takes the camera parameters and the detected boxes in each frame to track the oranges in the frame sequence. A 3-D relocalization component deals with occlusions and fruits exiting and entering the camera FOV. The bounding boxes are grouped in *tracks*, defining the path that every single orange performed in the frame sequence. These tracks are not necessarily *contiguous*, properly dealing with long-term occlusions and fruit reappearance. The

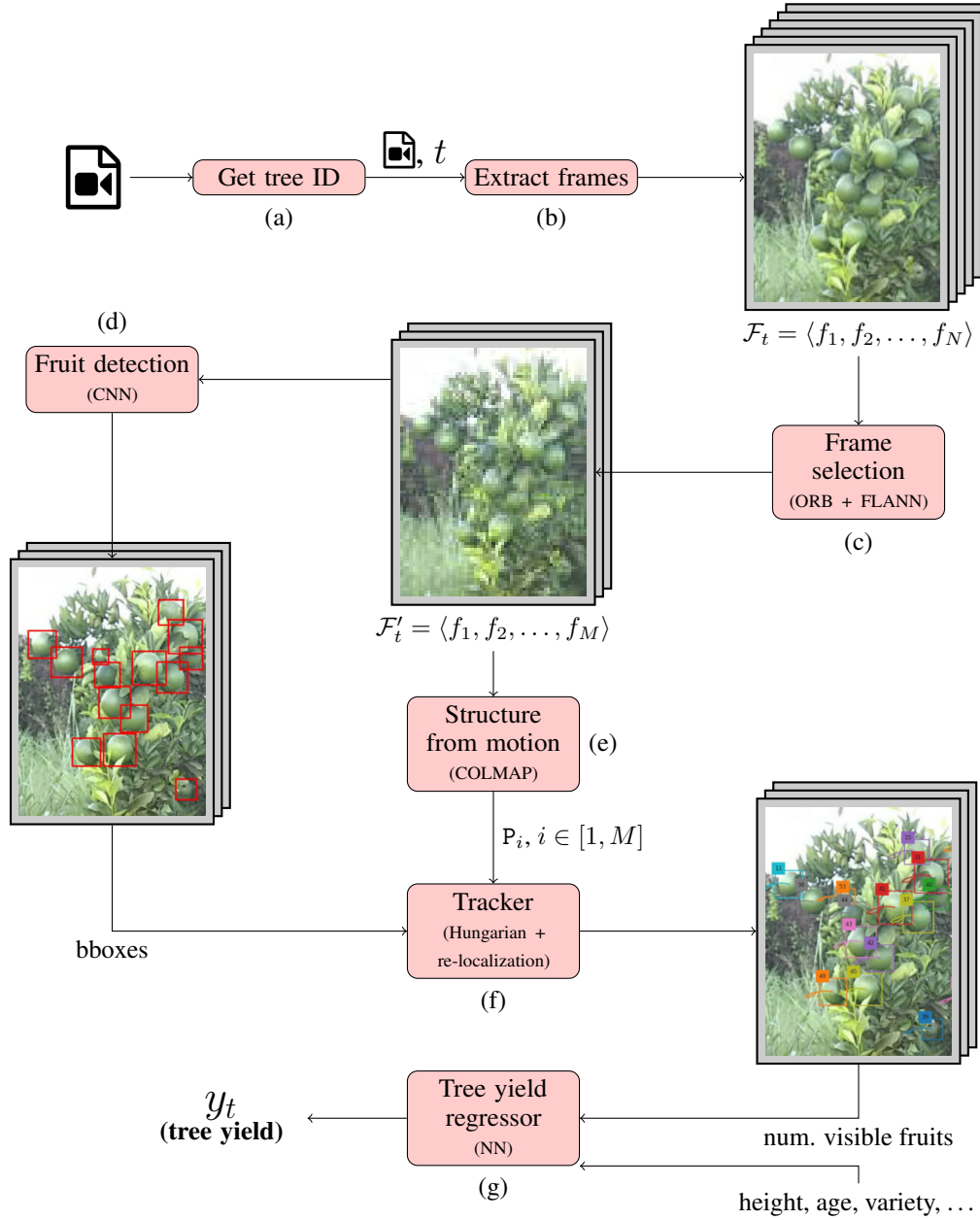


Figure 1: Pipeline overview. In the input video (single face of a single tree), a marker identifying the tree t is visible in the first frames. The tree is identified (a), and its frame sequence $\mathcal{F}_t$ is extracted (b). A frame selection procedure removes redundant frames (few or no motion), producing a reduced sequence $\mathcal{F}'_t$ (c). Fruit detection by a CNN (d) and Structure from motion by COLMAP (e) are performed for the frames in $\mathcal{F}'_t$. The ego-motion (set of projective matrices P_i) and the found bounding boxes feed the tracker (f). Finally, the number of found visible fruits and other tree data (height, age, ...) are employed by a regressor to estimate the total number of oranges in tree t (g).

number of tracks corresponds to the number of observed fruits in the video. Finally, this number is combined with the value found for the other face of the tree and with additional data, such as tree variety, height, and plant age. Such data is inputted to a yield regressor (Section 3.4), that produces a final estimation of the number of fruits y_t in the tree t .

3.1 Dataset

The pipeline shown in Figure 1 was executed for a set with more than 1,500 sweet orange trees. However, the computer vision-based components need annotated datasets for training and evaluation. Such annotation effort, although favored

by the new annotation tool presented in Section 3.1.5, could not be performed for the entire video set. A subset of 12 videos was annotated for multiple object tracking evaluation, forming the MORANGET dataset. Part of these annotations were combined to previously annotated images to form another dataset, ORANDET, built for orange detection. Such data allowed the training and evaluation of the fruit detection and tracker modules in the pipeline.

Although it was not feasible to evaluate multiple object tracking with all 1,543 trees, the whole set was processed through the pipeline and forwarded to the yield regressor. This last step, however, processed only 1,139 trees, which satisfied the requirement of having at least a fruit counting for one of its sides, as well as other information considered relevant, as one can see in Section 3.4.1. The following sections provide a more in-depth description of such data.

3.1.1 Orange Crop Forecast data

The Orange Crop Forecast, conducted by Fundecitrus [2022], utilized a stratified random sampling technique to select a representative sample of 1,560 orange trees. The initial drawing involved 1,200 trees that were proportionally distributed across the citrus belt, stratified based on their region, variety, and age. An additional drawing included 360 resets of younger ages to replace trees lost due to diseases, and other causes. To obtain the sample, a fruit stripping method was employed, involving the advanced harvest of all fruits from each tree. During the stripping process in the field, fruits from different flowering periods and consequently, in different phenological stages were sampled: green fruit close to the final size (classified as fruit from the 1st flowering period, $F1$); green fruit with a table tennis ball size (~ 4.5 cm diameter, classified as fruit from the 2nd flowering period, $F2$); and green fruit with a marble ball size (up to ~ 3.0 cm diameter, classified as from the 3rd period, $F3$) [Barbasso et al., 2005]. The fruit-stripping operation took place between March 28 and May 11, 2022, and the harvested fruits were transported to a laboratory in Araraquara, SP, Brazil. At the laboratory, the fruits were sorted into different bloom categories and quantified using automatic counting equipment before being weighed. Fundecitrus provided the ground-truth for fruit counting, tree height, age, and fruit variety for 1,543 trees.

3.1.2 Video recording

The crop forecast staff was instructed to record video sequences for each tree before the manual fruit stripping. Smartphones were employed to record videos presenting a resolution of 1940×1080 pixels. The staff was oriented to record the videos in portrait orientation, maximizing the field of view in the vertical direction. Considering the trees' density in commercial farms, it is not feasible to cover an entire tree canopy in a single pass, considering the employed smartphones do not present wide-angle lenses. So, for each tree in the sample, the forecast staff recorded a video for each side of that tree, i.e., the facade facing a row of the orchard. For each side, the staff member performed a trajectory, recording first the bottom of the tree, then moving to the central part and, finally, recording the top of the canopy, all contained in a single video take, as shown in Figure 2.

The staff works on a rigid schedule determined by the forecasting deadline, so the video sequences must be recorded under the available light conditions at that moment. If the light conditions are not ideal, for example, when the sun is behind the tree of interest, the staff member has to record the video anyway. This commonly produces frames presenting direct sunlight over the camera for the higher parts of the canopy. For the same reason, the video set presents diverse exposures, caused by different sunlight and shadows patterns in the trees.

3.1.3 Frame sampling

For each video (single tree side), all frames are initially extracted using the FFmpeg library⁷, producing a sequence of frames $\mathcal{F} = \langle f_1, f_2, \dots, f_N \rangle$ (see Figure 1). However, not all the N frames are needed for the fruit detection and tracking: similar neighboring frames can be removed to reduce processing time. The method employed for frame sampling is divided into two steps. Firstly, the ORB keypoint detector and descriptor [Rublee et al., 2011] is employed to extract features for each frame. After the first frame f_1 is added to the sample, the following procedure is iteratively repeated. Let f_i be the last frame in $\mathcal{F}$ included in the sample. Descriptors of the frame f_i are compared to those of a frame f_j , $j \in [i + 1, N]$ using the FLANN algorithm [Muja and Lowe, 2014], that produces a set of matching features between the two frames. If the number of corresponding descriptors is below a threshold, that indicates significant differences between frames, and f_j is included in the sample. Otherwise, the spatial coordinates are used to calculate the average distance between matched keypoints. If this average distance exceeds a defined threshold (10 pixels), f_j is added to the sample. This process is repeated between the last selected frame and the subsequent frames until all the original N frames are analyzed. After sampling, we have a sequence $\mathcal{F}' = \langle f_1, f_2, \dots, f_M \rangle$, $M < N$ (the remaining frames are sequentially re-indexed). On average, this sampling procedure reduces the number of frames by 50 to 70 percent.

⁷FFmpeg – <https://ffmpeg.org>.

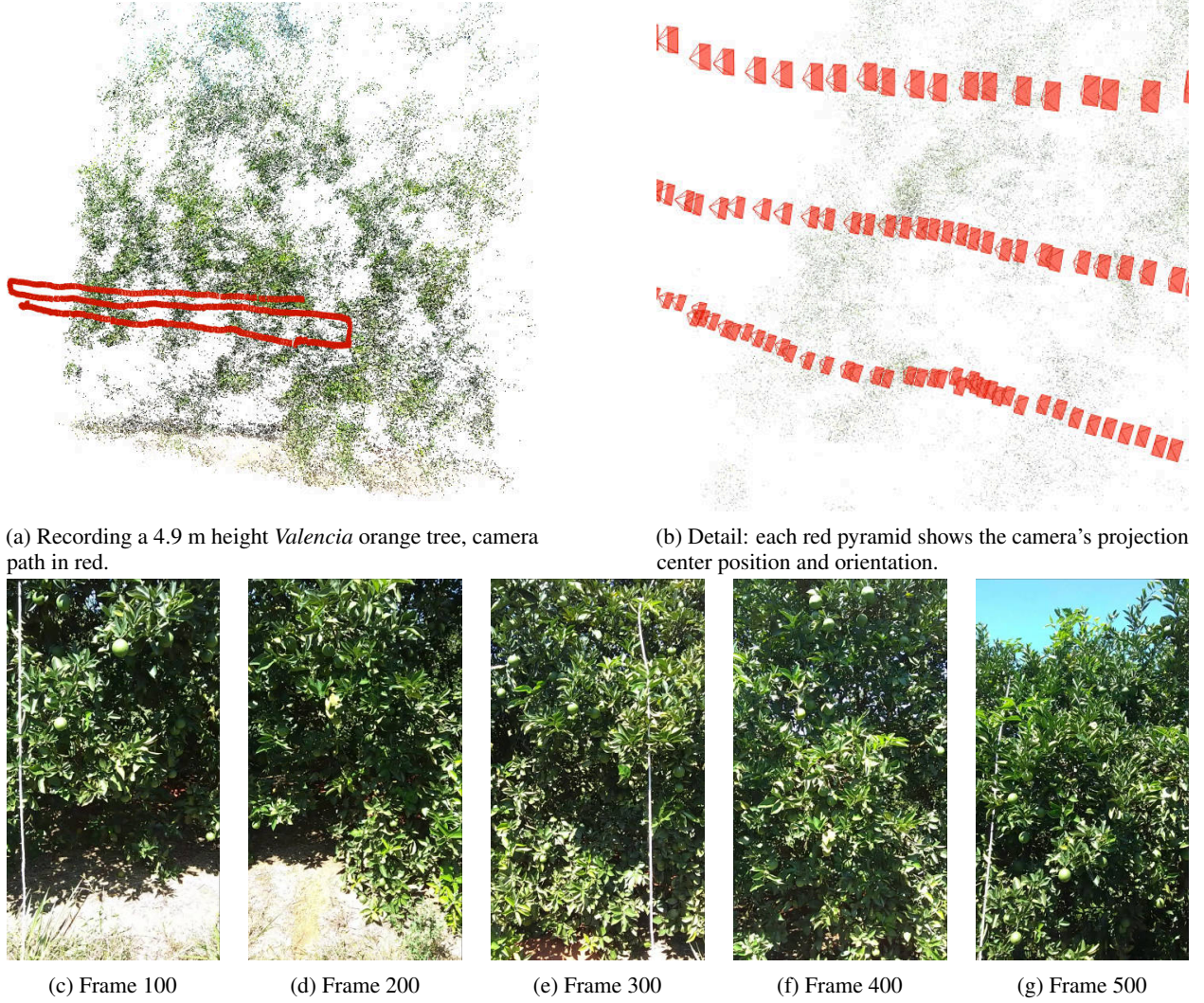


Figure 2: The tree recording process. A smartphone is used to record a video, following the path showed in red (a). The operator starts imaging the tree bottom, then following to the middle and top of the canopy, changing the camera orientation as needed (b). A sample of five frames is also shown (c–g).

3.1.4 Camera matrix estimation

Both our annotation tool and our fruit relocalization procedure rely on knowledge about the *camera projective matrix*. This matrix encodes all geometrical information needed to model the projective process, including the camera intrinsic parameters (focal distance, for example) and the extrinsic parameters: the camera position and orientation in the tridimensional space (ego-motion) [Hartley and Zisserman, 2004]. For each frame $f_i \in \mathcal{F}'$, we need a 3×4 projective matrix P_i such that:

$$\mathbf{x}_i = P_i \mathbf{X}, \quad (1)$$

where $\mathbf{X}$ is a tridimensional point in the scene and $\mathbf{x}_i$ its projection on frame f_i . The P_i matrices can be estimated from image sequences by pose graph optimization (PGO), simultaneous localization and mapping (SLAM), structure from motion (SfM) [Dellaert, 2021] or visual-inertial navigation (VIN) [Huang, 2019]. In this work, we have employed structure from motion, using the COLMAP tool proposed by Schonberger and Frahm [2016]. Figure 2a and Figure 2b show an example of the camera position and orientation recovered by COLMAP for a frame sequence in the dataset, and includes some tree *structure*: several points $\mathbf{X}$ in the scene that acted as landmarks in the SfM process [Dellaert, 2021]. The SfM results from COLMAP are also included in the public dataset.

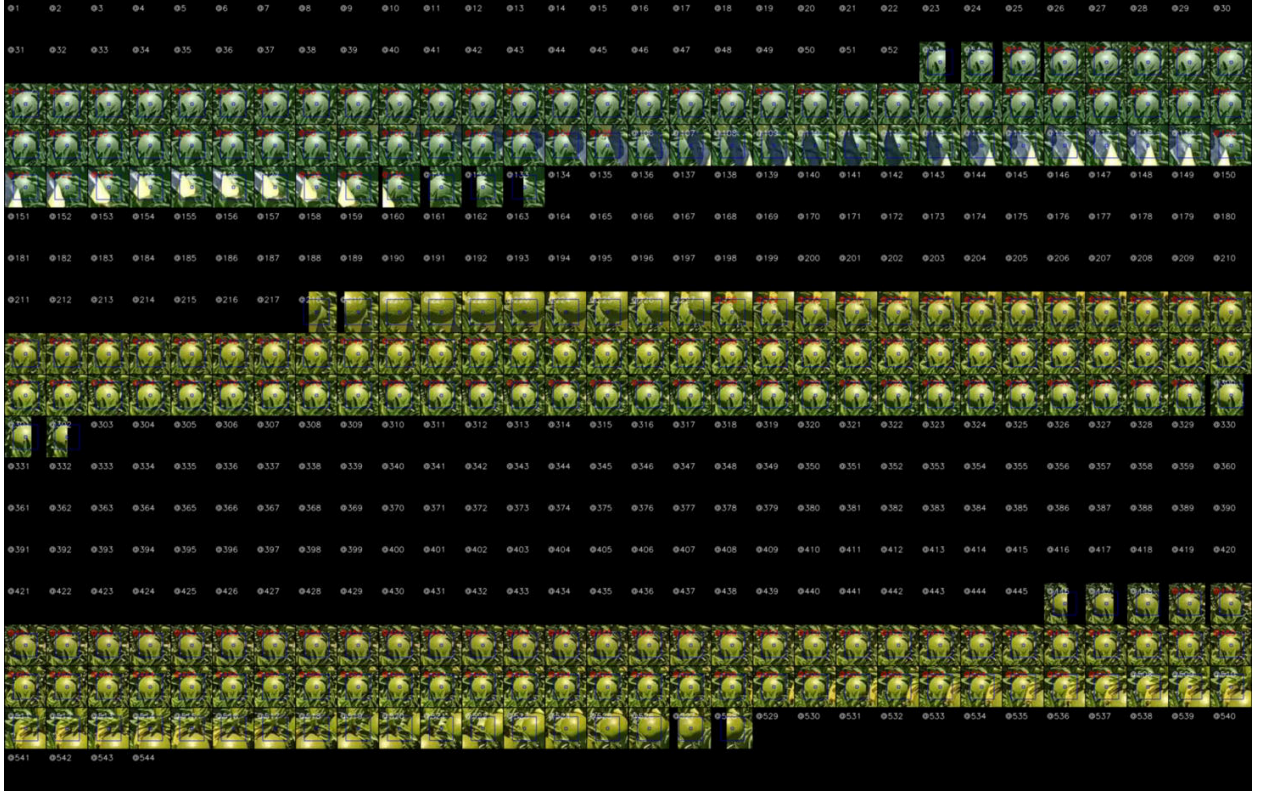


Figure 3: The mosaic window from the custom annotation tool for orange tracks annotation. The sequence is composed by 544 images (frames), represented by 544 squares in the mosaic. When the orange is in the FOV, a view extracted from the corresponding frame is displayed in the mosaic. White marks mean occlusion, while the red ones means above 50% visibility (non-occluded). Best seen in digital format.

3.1.5 Annotation and the MORANGET dataset for fruit tracking

As pointed by de Jong et al. [2022], multiple fruit tracking involves *homogeneous objects*. Identifying fruits that have been hidden from view for extended periods, either due to occlusions or being out of sight, is a challenging error-prone task, even for humans. Even when the fruit is visible, registering a bounding box for an orange in each frame is a slow and tedious work. Considering these issues, we have developed a new annotation tool that exploits the spherical shape of oranges and employs camera position information estimated by structure from motion.

The developed tool lets the users in charge of annotations to draw square bounding boxes for an orange in a few frames. Using the camera projective matrices P_i for each frame, the orange’s center and ray are estimated *in the 3-D space*. The fruit is reprojected on *all* frames (using Equation 1), being the tool able to automatically check if the fruit is in the FOV of each frame and, in the positive cases, draw a proper bounding box. The user can adjust every bounding box in every frame and, if needed, re-estimate the spherical orange in 3-D and reproject it again, refining the fruit localization. Nonetheless, the user is required to search for instances of occlusion where the orange is obstructed by leaves, branches, or other fruits, even though it is within the camera’s view. This method has made the process of annotating orange tracks faster and more reliable, especially when faced with challenging relocalization instances. The capability to modify the orange’s ray in 3-D and then project it onto frames results in well-adjusted bounding boxes, which would have otherwise required laborious efforts from annotators.

Figure 3 shows the *mosaic window* from the custom annotation tool. In this example, the tool is displaying the views of a single orange in a 544 frame long sequence. Each square in the mosaic corresponds to the view of that orange in the current frame, including part of its neighborhood. The orange’s center and bounding box are displayed in blue. Note the same orange enters and exits the field of view *three times*, entering at frames 53, 218 and 446. Furthermore, three occluded segments are observed: between frames 106–119, 218–227 (occlusion by other orange) and 508–528. We have adopted two conventions: if more than 50% of the fruit is visible, it is considered non-occluded (marked in red in Figure 3). Otherwise, the orange is considered occluded (marked in white). Moreover, if any part of the bounding box is out FOV, the orange is considered non-visible, to avoid unfairly penalizing trackers in borderline cases.

Table 1: MORANGET frame sequences. Each sequence corresponds to one side of one tree from five different regions in São Paulo state: Limeira (LIM), Duartina (DUA), Porto Ferreira (PFE), Brotas (BRO) and Avaré (AVA).

Sequence	Gimble	Region	Variety	Planting year	Height (cm)	Num. frames	Num. oranges
V01	No	LIM	<i>Pera</i>	2007	345	408	110
V02	Yes	LIM	<i>Pera</i>	2012	250	409	46
V03	Yes	DUA	<i>Valencia</i>	2005	490	634	90
V04	No	DUA	<i>Valencia</i>	2012	410	384	105
V05	No	PFE	<i>Natal</i>	2017	200	293	148
V06	Yes	BRO	<i>Valencia</i>	2009	400	447	122
V07	Yes	BRO	<i>Valencia</i>	2006	414	553	154
V08	Yes	DUA	<i>Valencia</i>	2013	320	544	192
V09	Yes	AVA	<i>Hamlin</i>	2019	183	222	16
V10	Yes	AVA	<i>Hamlin</i>	2019	183	244	13
V11	Yes	AVA	<i>Hamlin</i>	2008	380	268	87
V12	Yes	AVA	<i>Hamlin</i>	2008	380	307	115

Table 2: ORANDET, a dataset for orange detection. The set includes more than 220,000 bounding boxes annotated for 57,812 images (tiles).

	Num. images	Num. bounding boxes
Training	20,662	89,318
Validation	3,434	11,198
Test	33,716	121,685
Total	57,812	222,201

The MORANGET [Santos et al., 2023] dataset is composed of 12 sequences, as shown in Table 1. Nine of them were recorded by smartphones connected to a gimbal for stabilization (Zhiyun Smooth 4, Guilin Zhishen Information Technology Co.). The set included trees of four orange varieties, *Valencia*, *Natal*, *Pera*, and *Hamlin*, presenting different ages and heights, from four different regions in the São Paulo citrus belt. Although some mature fruits, most of the oranges are green: the crop forecast program needs to perform the assessments at an early stage of the fruits’ development. Note that the MORANGET dataset is employed to evaluate multiple fruit tracking and the counting of visible fruits in videos of single trees (single side), while the Orange Crop Forecast data is used for the evaluation of yield prediction for the 1,543 trees set.

3.1.6 ORANDET dataset for fruit detection

Utilizing the camera’s pose information enables the annotation of an extensive array of bounding boxes within the frames of videos in the MORANGET dataset. We have selected four sequences, V04, V05, V07 and V11 (Table 1) to compose a training set for orange detection. Each frame underwent a process of extracting a set of tiles, each measuring 416×416 pixels, which were then incorporated into the training set. Given the relatively diminutive size of fruits compared to frame dimensions, the application of tiling emerged as a straightforward yet efficacious strategy to mitigate the risk of overlooking small objects. This process resulted in a compilation of 21,031 tiles extracted from these frame sequences, with an additional 3,065 tiles incorporated from a previous study on orange detection [Camargo Neto et al., 2019]. Subsequently, the dataset was partitioned into training and validation subsets. The eight remaining frame sequences were exclusively designated for a test set, comprising 33,716 tiles. Table 2 outlines the definitive composition of the orange detection dataset, denominated as ORANDET, with illustrative examples depicted in Figure 4.

Within the context of the pipeline illustrated in Figure 1, the ORANDET dataset [Santos et al., 2022] was instrumental in both training and evaluating CNN-based fruit detection. Our orange tracker, depicted in Figure 1 (f) and expounded upon in Section 3.2.2, does not use any learning-based component. Consequently, the MORANGET dataset was exclusively employed to *evaluate* the tracking performance. Finally, the yield regression employs the Orange Crop Forecast data and the fruit counting from tracking to train and evaluate the yield predictions, as seen in Figure 1 (g).



Figure 4: Examples in the ORANDET training set. Images present immature green fruits. Note the variety in light, shadows, size, and blur.

3.2 Orange detection

We trained and evaluated six different architectures for orange detection: YOLOv3 [Redmon and Farhadi, 2018], YOLOv6 [Li et al., 2022], YOLOv7 [Wang et al., 2022], EfficientDet [Tan et al., 2020], and the YOLOv5 and YOLOv8 models proposed by Ultralytics⁸. Nowadays, there are several neural network architectures for object detection, based on CNNs [Arkin et al., 2023] and transformers [Shehzadi et al., 2023], and the selection of architectures is not straightforward. Considering the extensive use of YOLO-based networks in fruit detection [Tian et al., 2019, Wang et al., 2019], specially in citrus detection [Camargo Neto et al., 2019, Mirhaji et al., 2021, Xu et al., 2023], we have selected YOLOv3, the last architecture proposed by Redmon and Farhadi [2018] (the authors that introduced YOLO),

⁸See <https://docs.ultralytics.com/models>.

and a few of recent YOLO-based approaches. To provide a comparative analysis, we have opted for EfficientDet as an alternative approach.

We used the implementations available in MMYOLO⁹ and MMDetection [Chen et al., 2019]. All models were trained using stochastic gradient descent (SGD) optimizer with momentum. Random flip augmentation was applied in the training pipeline for all models, and the Mosaic augmentation, introduced after YOLOv3, was used during the training of YOLOv5, YOLOv6, YOLOv7, and YOLOv8.

3.3 Multiple orange tracking

Our multiple fruit tracking method gets as input the bounding boxes set $\mathcal{B}_i$ from the orange detector and the projective matrices P_i for each frame f_i in the frame sequence $\mathcal{F}_t'$ (see Figure 1). The method is based on a few components, described as follows.

3.3.1 Bounding box assignments

An essential component of the method is data association: *bounding box assignment* is performed between two box sets, matching boxes in a set $\mathcal{B}_i$ to boxes in another set $\mathcal{B}_j$. Such association is performed by a variant of the Hungarian algorithm Kuhn [1955], using a cost function based on intersection over union (IoU):

$$\text{cost}(\mathbf{b}_i^{(p)}, \mathbf{b}_j^{(q)}) = 1 - \text{IoU}(\mathbf{b}_i^{(p)}, \mathbf{b}_j^{(q)}), \quad (2)$$

where $\mathbf{b}_i^{(p)} \in \mathcal{B}_i$, $\mathbf{b}_j^{(q)} \in \mathcal{B}_j$, and IoU is the *Jaccard index* defined over the intersection area and union area between $\mathbf{b}_i^{(p)}$ and $\mathbf{b}_j^{(q)}$:

$$\text{IoU}(\mathbf{b}_i^{(p)}, \mathbf{b}_j^{(q)}) = J(\mathbf{b}_i^{(p)}, \mathbf{b}_j^{(q)}) = \frac{|\mathbf{b}_i^{(p)} \cap \mathbf{b}_j^{(q)}|}{|\mathbf{b}_i^{(p)} \cup \mathbf{b}_j^{(q)}|}. \quad (3)$$

Considering P bounding boxes in $\mathcal{B}_i$ and Q boxes in $\mathcal{B}_j$, the assignment matrix $\mathbf{A}$ is a $P \times Q$ matrix where $\mathbf{A}[p, q] = 1$ iff $\mathbf{b}_i^{(p)}$ and $\mathbf{b}_j^{(q)}$ are associated, otherwise $\mathbf{A}[p, q] = 0$. Assume, without loss of generality, that $P \leq Q$. The association problem is a *2-D rectangular assignment* minimization problem [Crouse, 2016]:

$$\mathbf{A}^* = \arg \min_{\mathbf{A}} \sum_{p=1}^P \sum_{q=1}^Q \text{cost}(\mathbf{b}_i^{(p)}, \mathbf{b}_j^{(q)}) \mathbf{A}[p, q], \quad (4)$$

subject to:

$$\sum_{q=1}^Q \mathbf{A}[p, q] = 1, \forall p \quad (5)$$

$$\sum_{p=1}^P \mathbf{A}[p, q] \leq 1, \forall q. \quad (6)$$

Note that $\mathbf{b}_i^{(p)}$ refers to the p -th bounding box in $\mathcal{B}_i$ and $\mathbf{b}_j^{(q)}$ refers to the q -th bounding box in $\mathcal{B}_j$. For this minimization, we have employed a modified Jonker-Volgenant algorithm with no initialization [Crouse, 2016], a $O(n^3)$ variant of the Hungarian algorithm. This implementation supports *unbalanced assignments*, so working in cases where $P \neq Q$: boxes in i not assigned (lost boxes) or boxes in j not assigned (novel boxes). As a post-processing step, we remove assignments where $\text{IoU}(\mathbf{b}_i^{(p)}, \mathbf{b}_j^{(q)}) = 0$ eventually produced by the minimization algorithm. The assignment component is repeatedly employed in this work:

- to perform data association between box sets $\mathcal{B}_i$ and $\mathcal{B}_{i+1}$, corresponding to detected fruits in two neighboring frames f_i and f_{i+1} ;
- to perform relocation, associating reprojected bounding boxes of previously seen oranges to boxes in $\mathcal{B}_i$, detected in the current frame f_i , and
- to evaluate results, matching detected boxes to the ground-truth annotation set $\hat{\mathcal{B}}_i$ for frame f_i .

⁹<https://github.com/open-mmlab/mmyolo>

3.3.2 Orange 3-D localization

The association procedure, when applied to bounding boxes sets from neighboring frames f_i and f_{i+1} , produces *contiguous tracks* in the form $\mathcal{T} = \langle \mathbf{b}_i, \mathbf{b}_{i+1}, \dots, \mathbf{b}_{i+n} \rangle$ (here the upper script indexes are omitted to simplify the notation). When a contiguous track is sufficiently long (we have adopted $n \geq 5$), we employ the camera matrices $P_i, P_{i+1}, \dots, P_{i+n}$ to estimate the three-dimensional position of the corresponding orange. Algorithm 1 takes as input a track $\mathcal{T}$ and the corresponding sequence of camera matrices $\mathcal{P}$ to estimate the center of the orange in the 3-D space, $\mathbf{X}$. The algorithm is based on two concepts: *reprojection error* and *random sample consensus*. The reprojection error is the distance between the bounding box center $\mathbf{c}_i$ (line 4) and $\mathbf{x}_i$, the reprojection of $\mathbf{X}$ on the i -th frame (Equation 1). A bounding box is considered an *inlier* iff this distance is within a threshold, the maximum reprojection error allowed (lines 15–16). The RANDOM SAMPLE CONSENSUS (RANSAC) is a general framework for robust estimation algorithms proposed by Fischler and Bolles [1981]: (i) a sample $\mathcal{S}$ is randomly selected to estimate the target; (ii) the points in the full set within a distance threshold are defined as inliers (consensus); (iii) the target is re-estimated using all inliers, and (iv) the process is iterated if the number of inliers is insufficient. In Algorithm 1, our sample $\mathcal{S}$ is a combination of three bounding boxes in the track (line 6) and a target $\mathbf{X}_\mathcal{S}$ is estimated using the *direct linear triangulation* procedure (DLT) [Hartley and Zisserman, 2004] (line 7). The reprojection is employed to define the set of inliers (lines 12–19) and, if the number of inliers is sufficient (line 20), $\mathbf{X}$ is estimated from the consensus set, again using DLT. However, considering consensus could not be achieved and the number of combinations of three boxes (line 6) can be massive, a maximum number of iterations is defined (lines 24–27). If the maximum number of iterations is reached or if all samples were considered with no consensus, the algorithm returns $\mathbf{X}$ as NIL and an empty inliers set (no success). In Algorithm 1, INDEXES($\cdot$) returns the frame indexes i for the bounding boxes $\mathbf{b}_i$ in the set (in a valid track, there is only one bounding box per frame).

Algorithm 1 RANSAC-based triangulation

```

1: procedure RANSACTRIANGULATION( $\mathcal{T}, \mathcal{P}$ )
2:   n_iters  $\leftarrow$  1
3:   for all  $\mathbf{b}_i \in \mathcal{T}$  do
4:      $\mathbf{c}_i \leftarrow \left( \frac{\mathbf{b}_i.x + \mathbf{b}_i.width}{2}, \frac{\mathbf{b}_i.y + \mathbf{b}_i.height}{2} \right)^\top$  ▷ Bounding box centroid
5:   end for
6:   for all  $\mathcal{S} \leftarrow \text{COMBINATIONS}(\mathcal{T}, 3)$  do ▷ RANSAC loop
7:      $\mathbf{X}_\mathcal{S} \leftarrow \text{DLTTRIANGULATION}((\mathbf{b}_j, P_j), j \in \text{INDEXES}(\mathcal{S}))$  ▷ Direct Linear Transformation (DLT)
8:     if  $\mathbf{X}_\mathcal{S}$  is NIL then
9:       continue
10:    end if
11:     $\mathcal{I} \leftarrow \emptyset$ 
12:    for all  $j \in \text{INDEXES}(\mathcal{T})$  do
13:       $\mathbf{x}_j \leftarrow P_j \cdot \mathbf{X}_\mathcal{S}$  ▷ Homogeneous form:  $\mathbf{x}_j = (x, y, z)^\top$ 
14:       $\mathbf{x}_j \leftarrow (x/z, y/z)^\top$ 
15:      geom_error  $\leftarrow \|\mathbf{x}_j - \mathbf{c}_j\|$  ▷ Reprojection error
16:      if geom_error  $\leq$  max_geom_error then
17:         $\mathcal{I} \leftarrow \mathcal{I} \cup \{\mathbf{b}_j\}$ 
18:      end if
19:    end for
20:    if  $\frac{|\mathcal{I}|}{|\mathcal{T}|} \geq \text{inliers\_ratio}$  then
21:       $\mathbf{X} \leftarrow \text{DLTTRIANGULATION}((\mathbf{b}_j, P_j), j \in \text{INDEXES}(\mathcal{I}))$  ▷ Re-estimate  $\mathbf{X}$  using all inliers
22:      return  $\mathbf{X}, \mathcal{I}$ 
23:    end if
24:    n_iters  $\leftarrow$  n_iters + 1
25:    if n_iters  $>$  max_iters then ▷ Exceeds the maximum number of iterations allowed
26:      return NIL,  $\emptyset$ 
27:    end if
28:  end for
29:  return NIL,  $\emptyset$  ▷ It could not find an acceptable estimation
30: end procedure

```

Algorithm 2 models the orange as a sphere and estimates the fruit’s center $\mathbf{X}$ and ray r_{3D} . The three-dimensional point $\mathbf{X}$ and the inliers bounding boxes set $\mathcal{I}$ are computed using Algorithm 1. Then the ray is estimated using the dimensions

of the boxes in $\mathcal{I}$ and similar triangles' ratio:

$$r_{3D} \approx \frac{r_i d}{f_{\text{focal}}} \quad (7)$$

where d is the distance between the camera center position $\mathbf{C}_i$ at frame f_i and the orange center $\mathbf{X}$ (line 10), f_{focal} is the camera focal distance, and r_i the 2-D ray estimated from the bounding box $\mathbf{b}_i$ dimensions (line 8). Each r_i provides an estimation for r_{3D} (line 11), and we use the median of the estimations set $\mathcal{R}$ as our final estimation for the orange three-dimensional ray. We have noted that the bounding boxes usually exceed the fruit boundaries¹⁰, overestimating the ray, then we multiply the median by a constant c for a better fit (line 13, we have adopted $c = 0.9$). The camera center $\mathbf{C}_i$ is easily computed from $\mathbf{P}_i$ (line 9) using:

$$\mathbf{C}_i = -\mathbf{M}^{-1} \mathbf{p}_4, \quad (8)$$

where $\mathbf{P}_i = [\mathbf{M} \mid \mathbf{p}_4]$, i.e., the $\mathbf{M}$ matrix is composed by the three first columns of $\mathbf{P}_i$ and $\mathbf{p}_4$ is the last column of $\mathbf{P}_i$.

Algorithm 2 Estimate orange's position and ray in 3-D

```

1: procedure ESTIMATEORANGE( $\mathcal{T}, \mathcal{P}$ )
2:    $\mathbf{X}, \mathcal{I} \leftarrow \text{RANSACTRIANGULATION}(\mathcal{T}, \mathcal{P})$  ▷  $\mathcal{I}$  is the set of inliers
3:   if  $\mathbf{X}$  is NIL
4:     return NIL, NIL
5:   end if
6:    $\mathcal{R} \leftarrow \emptyset$ 
7:   for all  $\mathbf{b}_i \in \mathcal{I}$  do then
8:      $r_i \leftarrow \max(\mathbf{b}_i.\text{height}, \mathbf{b}_i.\text{width})/2$  ▷ Estimate 2-D ray from bounding box dimensions
9:      $\mathbf{C}_i \leftarrow \text{GETCAMERACENTER}(\mathbf{P}_i)$  ▷ Camera projection center
10:     $d \leftarrow \|\mathbf{X} - \mathbf{C}_i\|$  ▷ Orange to camera distance
11:     $\mathcal{R} \leftarrow \mathcal{R} \cup \left\{ \frac{d \cdot r_i}{f_{\text{focal}}} \right\}$ 
12:  end for
13:   $r_{3D} \leftarrow c \cdot \text{median}(\mathcal{R})$ 
14:  return  $\mathbf{X}, r_{3D}$ 
15: end procedure
```

In our MOT tracking for oranges, tracks can be *discontinuous*, meaning that the fruit is not visible at some frames because it is occluded or out of view. Note that Algorithm 1 and Algorithm 2 have no need for continuous tracks: they can operate on discontinuous ones as new bounding boxes are added to the tracks. Tracks can present one of two possible states in our tracking system: LOST, if the track has no bounding box in the current frame f_i , and ACTIVE, if the track presents an observed box $\mathbf{b}_i$. When processing the current frame f_i , our system starts performing *relocalization*, using the Hungarian association to match LOST tracks to bounding boxes in $\mathcal{B}_i$ yet not associated to any track.

3.3.3 Relocalization

The LOST tracks have their fruits' position and ray estimated by Algorithm 2: a center $\mathbf{X}$ and a ray r_{3D} are available for each track. Using Equation 1, we find the reprojection of the orange's center $\tilde{\mathbf{x}}_i = \mathbf{P}_i \mathbf{X}$ at frame f_i . The ray r_{3D} and Equation 8 are employed to find the ray $\tilde{r}_i$ for the orange on f_i . Finally, the reprojected bounding box is defined as:

$$\tilde{\mathbf{b}}_i = [\tilde{x}_i - \tilde{r}_i, \tilde{y}_i - \tilde{r}_i, 2\tilde{r}_i, 2\tilde{r}_i].$$

where $\tilde{\mathbf{x}}_i = (\tilde{x}_i, \tilde{y}_i, 1)^\top$ (homogenous coordinates) and $\tilde{\mathbf{b}}_i$ is the *reprojected bounding box* for the track in frame f_i (the $\tilde{\cdot}$ notation is used to indicate the values came from the orange *estimation*, not from orange *detection*). Such boxes are computed for all LOST tracks, producing a set of boxes $\tilde{\mathcal{B}}_i$. The Hungarian association (Equation 4) is employed to match boxes from LOST tracks in $\tilde{\mathcal{B}}_i$ to boxes in $\mathcal{B}_i$ yet not matched to any track. In the case of a track is successfully associated to a box, its state is changed to ACTIVE.

3.3.4 Next frame association

After relocalization, the tracking system will employ the Hungarian association to match boxes $\mathbf{b}_i^{(p)} \in \mathcal{B}_i$ to boxes $\mathbf{b}_{i+1}^{(q)} \in \mathcal{B}_{i+1}$. If the association algorithm matches $\mathbf{b}_i^{(p)} \rightarrow \mathbf{b}_{i+1}^{(q)}$, and $\mathbf{b}_i^{(p)}$ belongs to an ACTIVE track $\mathcal{T}$, the box

¹⁰Note also Equation 7 is an approximation: even in the absence of noise, the equality would just hold if the sphere center was perfectly projected in the image's principal point [Hartley and Zisserman, 2004].

$\mathbf{b}_{i+1}^{(q)}$ is appended to $\mathcal{T}$. Otherwise, a novel track $\mathcal{T}' = \langle \mathbf{b}_i^{(p)}, \mathbf{b}_{i+1}^{(q)} \rangle$ is created. After association, tracks not matched to any box $\mathbf{b}_{i+1}^{(q)}$ change to the LOST state, while all the remaining ACTIVE tracks presenting more than 5 bounding boxes have their oranges re-estimated using Algorithm 2. The entire procedure (relocalization followed by next frame association) is repeated for each frame until f_{M-1} .

In summary, contiguous ACTIVE tracks presenting at least 5 boxes initialize 3-D spherical models for oranges, parameterized by central point and ray. After this initialization, the fruit is passive of relocalization: if the bounding boxes' association fails by any reason, producing LOST tracks, the reprojection of the spherical model on the current frame can be employed to relocalize the fruit, associating the reprojected box to a detected one. After relocalization, the track is ACTIVE again and can be updated with new detected boxes. Such track is now discontinuous, registering where (bounding box) and when (frame) the fruit is visible. Furthermore, the orange model (central point and ray) is continuously refined as new detections are added to the track. Only the tracks with successfully estimated 3-D orange models are considered for fruit counting.

3.3.5 Implementation

The tracker was implemented in Python 3. The 2-D rectangular assignment implementation employed is from SciPy's optimization module [Crouse, 2016]. The DLT algorithm was implemented using singular value decomposition [Golub and van Loan, 2013] available in SciPy's linear algebra module.

3.3.6 Tracking evaluation

Villacrés et al. [2023] presented a *sensitivity analysis* aimed to evaluate the performance of a fruit tracker under varying probabilities of fruit detection. The analysis begins with perfect detection, from the ground-truth data, where all fruits are identified accurately, and then progressively eliminates certain bounding boxes. Each bounding box in the ground truth is treated as a random variable with a uniform distribution within the range $[0, 1]$ for each bounding box in f_i . Based on the value, each bounding box is kept or removed from the detections set $\mathcal{B}_i$. Specifically, if the random value of a fruit is below or equal to a pre-defined threshold, it is considered detected; otherwise, it is discarded. Similarly to Villacrés et al., in the present work we evaluated different values of the probability of detection, using 0.4, 0.6, 0.8, and 1 as thresholds (a threshold of 1 equals to "perfect detection", a copy of ground-truth bounding box data). The tracker is also evaluated using the detections from the CNN-based orange detection module presented in Section 3.2. The evaluation present values for HOTA [Luiten et al., 2021] and MOTA [Bernardin and Stiefelagen, 2008]. We also include in the evaluation two components of HOTA: DetA, the percentage of aligning detections, and AssA, the average alignment between matched trajectories. Appendix A presents mathematical definitions for HOTA, DetA and AssA, adapting the original Luiten et al. [2021] formulation to the notation adopted in this work.

3.4 Yield regressor

After the fruit detection and tracking stages, we have all the fruits in a tree identified and mapped in 3-D. However, some fruits may have not been identified or tracked due to problems in video acquisition or because they are located inside the canopy. To address this problem, we propose a regressor that takes the number of fruits on each side of the plant as input and estimates the actual number of fruits on that plant. This estimator can be based on statistics, as proposed in [Koirala et al., 2019a], in which the authors consider statistical differences in each orchard to propose a correction factor. Alternatively, it can be implemented as a machine learning model. In our case, we use a machine learning model that learns an implicit representation accounting for the impact of other variables, such as variety, group of variety, dimensions of the plant, region, and sector, to propose an estimated number of fruits.

3.4.1 Data used for regression

Although we received raw data for 1,543 trees, only 1,197 trees could be successfully processed in the previous stages of the pipeline. Trees with ill-recorded videos fail in the SfM part of the pipeline, and do not follow through the tracking and the yield detection steps. Table 3 shows the first four lines of a regression data table. Tracking generated the last two columns, *CbyT-A* and *CbyT-B* (*counting by tracking*), which correspond respectively to the fruits automatically counted on sides A and B of the tree. These counting columns can change according to the tracker employed, for example, a tracking using YOLOv3-based detections or counting results from a YOLOv5-based one. Such counting values were aggregated to the original data to enrich the information provided to the regressor. However, not all 1,197 records could be used for training the regressor, as we had to exclude plants presenting missing data and those without fruit counting from tracking for either side.

Table 3: Sample containing the first four lines of the dataset used for training and testing of the regressor. ID corresponds to the plant identifier; Sector and Region are references for geographical location; VarG and Var refer to the variety group and variety of the plant, respectively; AG is a classification for the age of the plant; $F1$, $F2$, $F3$ and $F4$ are the number of fruits manually counted in sizes corresponding to the first to the fourth flowering; H , W and D are the dimensions of the plant; CbyT-A and CbyT-B are the number of visible fruits automatically counted using the vision-based pipeline for both sides of the plant.

ID	Sector	Region	VarG	Var.	AG	$F1$	$F2$	$F3$	$F4$	H	W	D	CbyT-A	CbyT-B
103	North	LIM	VT	<i>Val.</i>	1	0	5	26	1	280	280	300	15	
1031	South	LIM	NT	<i>Nat.</i>	3	0	5	54	10	290	360	295		19
106	North	DUA	PRME	<i>Pera</i>	2	0	0	58	36	340	300	335	10	10
104	North	DUA	PRME	<i>Pera</i>	3	400	0	11	0	310	460	495		32

3.4.2 Data preprocessing

Before submitting data to the model, it is necessary to perform some operations to adequate these data to the dynamics of the processing of the artificial neural network. The following operations were executed over the dataset:

Transformation of categorical variables: neural networks only accept numerical values as inputs to their neurons, which means that categorical variables must be transformed into numerical values. For example, the value *Hamlin*, which is one of the possible values for the variable VARIETY, has to be converted to a number. One of the possible methods is using One Hot Bit Encoding [Géron, 2019], which creates a new column for each possible value of the variable. The column *Hamlin*, for instance, receives value 1 for whenever a sample is classified with that variety. This procedure is executed for all categorical variables.

Standardization of numerical values: to facilitate convergence to the optimal training point, numerical values should be standardized. Variables with very discrepant scales, such as one ranging from [0–100,000] and another from [-1.0–1.0], can cause problems because the gradients calculated during back propagation will have very different effects when applied to different values. One solution for this problem is to apply standardization using the formula $\mathbf{u}_i = (\mathbf{v}_i - \bar{\mathbf{v}}) / \sigma_{\mathbf{v}}$, where vector $\mathbf{u}_i$ contains the standardized values of $\mathbf{v}_i$, $\bar{\mathbf{v}}$ is the mean value of the vectors $\mathbf{v}_i$ and $\sigma_{\mathbf{v}}$ its standard deviation.

3.4.3 Machine learning regression model

We have tried several machine learning algorithms for the regression problem, such as Support Vector Machines [Cortes and Vapnik, 1995], Bagging [Breiman, 1996] and Gradient Boosting [Friedman, 2001]. None of them, however, had a better performance than a multilayer feed-forward neural network [Goodfellow et al., 2017].

Neural networks are highly flexible in their assembly. For instance, one can create a neural network with just a single layer, and it still is able to solve a problem. However, tasks such as image classification usually requires dozens or even hundreds of layers. This flexibility is also a weakness, as it necessitates searching through a wide range of configurations to find the most suitable one for a specific problem [Géron, 2019]. A common heuristic approach is to gradually increase the number of layers and neurons and evaluate how the results converge towards the desired outcome. Once the parameter range with the best results is identified, it can be further refined. In the case of the yield regressor, the neural network should not be excessively complex due to the amount of data available. If a neural network has too many parameters, it may suffer from overfitting, where it learns details in the training dataset and performs poorly with the test data as compared to its training performance.

In our path to find the best network configuration, we relied on *cross validation*, a widely used technique in machine learning, to statistically assess whether one computational experiment performs better than another [Kohavi, 1995]. Cross validation involves running the same algorithm multiple times, allowing for more reliable performance statistics and a certain level of confidence. In our research, we divided the training dataset into ten parts, also known as folds. The process entails training the algorithm on nine folds while reserving the tenth fold for evaluation. This process repeats until all ten folds have been used as the test set. To determine the level of certainty regarding the superiority of a model, we employed a *statistical hypothesis test*. Since we had ten folds, the assumption that the data follow a normal distribution does not hold true. The normal distribution is typically reserved for datasets larger than 30 elements with known variance [King and Zisserman, 2019]. Instead, we used the t-Student distribution for our set of ten performance measurements. We set a p-value of 0.05 to ensure that we select a model with 95% confidence.

Table 4: Feed-forward neural network architecture employed on yield regression. B corresponds to the batch size, i.e., the number of data entries.

Layer (type)	Output Shape	Num. Params
dense_1 (Dense)	$(B, 7)$	294
batch_normalization_1	$(B, 7)$	28
dense_2 (Dense)	$(B, 7)$	56
dense_3 (Dense)	$(B, 1)$	8
Total params: 386		
Trainable params: 372		
Non-trainable params: 14		

We conducted experiments with neural networks comprising one to six layers, and varying the number of neurons per layer from seven to 28. The last layer of the regressor consists of a single neuron, which is responsible for the calculation of the yield estimation. Among the tested architectures, the one presented in Table 4 demonstrated the best results, as will be discussed in the next section. The model was implemented in Python, using the Keras API¹¹, which is built on top of the TensorFlow™ platform.

4 Results

4.1 Tracking sensitivity to detection

In the sensitivity analysis, tracking is performed by the algorithms presented in Section 3.3, with no component based on machine learning: all boxes come from the ground-truth, with random removal of some boxes to emulate detection components presenting different detection rates. Table 5 shows HOTA, DetA¹² and AssA results for four different rates. The table also displays the MOTA metric for comparison, considering that a few works [de Jong et al., 2022, Villacrés et al., 2023] employed that metric. The *CbyT* (counting by tracking) value is the fruit counting, corresponding exactly to the number of (possibly discontinuous) tracks found by our MOT system, while *CbyT-GT* is the ground-truth value. The last column displays the relative error: the ratio of absolute error to the ground-truth value.

Perfect fruit detection does not imply in perfect tracking and counting: association errors, as seen in Figure 13, induced by occlusions, cause tracking errors. However, the tracking performance in high (HOTA and MOTA above 0.9), resulting in accurate counting: 2.34% relative error in counting of visible fruits for a 100% detection rate (considering individual trees, the median for the relative error is 1.23%). But accurate counting can be reached even under imperfect detections: an 80% detection rate could reach a 3.01% relative error (median 2.20%). As expected, low detection rates will severely damage counting: a 60% detection rate implied in a 14.44% error in counting (median 12.35%), while a 40% rate reached an error of 55.93% (median 55.77%).

Figure 5 shows tracking examples for two videos, V07 and V12. The tracking was performed assuming an 80% detection rate. When the orange position and ray were estimated by Algorithm 2, the numeric ID for the track (fruit) is displayed. Tracks not yet presenting a successful estimation are marked with a NIL label, otherwise. LOST tracks are displayed as white, dashed boxes. Note as, at each frame, there is a significant number of LOST tracks, but the system keeps projecting them on each frame at their expected locations. A relocalization example can be seen for track 94 in V07: the track is lost at frames f_{260} and f_{265} because of occlusion, but the track became ACTIVE at f_{270} . Examples of new tracks creation and orange estimation can be seen for fruits 95, 96 and 97, again in video V07. The three-dimensional positions estimated for each orange in V07 are shown in Figure 6.

The importance of relocalization is highlighted in Table 6, which presents sensitivity results for both 100% and 80% detection rates, but without incorporating relocalization. In the case of multiple orange tracking, effectively managing occlusions or re-entering fruits is crucial. Without proper handling of these scenarios, the generated counting becomes unstable and unusable, as observed in the relative error column. A comparison between the results in Table 5 and Table 6 emphasizes the necessity of long-term tracking for accurate counting. Even a more advanced tracker that incorporates motion estimation would struggle to handle occlusions and re-entering objects without a dedicated long-term association component [Bewley et al., 2016].

¹¹<https://keras.io/about>.

¹²Note that, for 100% detection rate, to see DetA values below 1.0 sounds counterintuitive: readers should consider that the assignments between detections and ground-truth are defined by the association matrices $\hat{A}^{(\alpha)}$ that *maximize* HOTA [Luiten et al., 2021], so association errors could produce some false positives and false negatives in Equation 20, see Appendix A.

Table 5: Sensitivity analysis. The table shows how tracking (and fruit counting) degrades with detection performance.

Detection (%)	Sequence	HOTA	DetA	AssA	MOTA	CbyT	CbyT-GT	Relative error (%)	Median
100%	V01	0.90691	0.90390	0.91079	0.96139	103	110	6.36%	1.23%
	V02	0.90277	0.90130	0.90683	0.97152	45	46	2.17%	
	V03	0.94512	0.94140	0.94907	0.97387	89	90	1.11%	
	V04	0.94159	0.94022	0.94306	0.98173	104	105	0.95%	
	V05	0.92378	0.93034	0.91765	0.95370	146	148	1.35%	
	V06	0.92984	0.92576	0.93423	0.97897	121	122	0.82%	
	V07	0.93595	0.92823	0.94428	0.97714	149	154	3.25%	
	V08	0.95137	0.94812	0.95515	0.99380	190	192	1.04%	
	V09	0.78204	0.62303	0.98163	0.63495	10	16	37.50%	
	V10	0.96471	0.96313	0.96640	0.99781	13	13	0.00%	
	V11	0.92858	0.90792	0.95016	0.95871	85	87	2.30%	
	V12	0.94809	0.94741	0.94898	0.99573	115	115	0.00%	
	All	0.93516	0.92761	0.94361	0.97308	1170	1198	2.34%	
80%	V01	0.68845	0.68009	0.69767	0.71980	101	110	8.18%	2.20%
	V02	0.69783	0.69759	0.70022	0.74322	47	46	2.17%	
	V03	0.72111	0.72464	0.71773	0.73939	88	90	2.22%	
	V04	0.72459	0.72899	0.72030	0.75362	104	105	0.95%	
	V05	0.67268	0.70495	0.64206	0.70454	147	148	0.68%	
	V06	0.72021	0.71768	0.72294	0.75514	119	122	2.46%	
	V07	0.71791	0.71040	0.72589	0.74529	144	154	6.49%	
	V08	0.72352	0.72368	0.72372	0.74971	189	192	1.56%	
	V09	0.63278	0.56457	0.70923	0.57254	12	16	25.00%	
	V10	0.76247	0.75657	0.76851	0.78134	13	13	0.00%	
	V11	0.69139	0.67515	0.70832	0.71356	82	87	5.75%	
	V12	0.72052	0.72119	0.72001	0.75194	116	115	0.87%	
	All	0.71210	0.71085	0.71386	0.73859	1162	1198	3.01%	
60%	V01	0.42052	0.37871	0.46725	0.39628	77	110	30.00%	12.35%
	V02	0.43747	0.42167	0.45582	0.44970	36	46	21.74%	
	V03	0.45983	0.46422	0.45555	0.46182	81	90	10.00%	
	V04	0.48905	0.49809	0.48025	0.49698	100	105	4.76%	
	V05	0.41860	0.43778	0.40030	0.41636	116	148	21.62%	
	V06	0.46982	0.45067	0.48991	0.46902	113	122	7.38%	
	V07	0.48637	0.47560	0.49763	0.49048	132	154	14.29%	
	V08	0.48001	0.47564	0.48464	0.48208	172	192	10.42%	
	V09	0.42851	0.38464	0.47741	0.38014	11	16	31.25%	
	V10	0.46489	0.48524	0.44543	0.47886	12	13	7.69%	
	V11	0.44513	0.41282	0.48018	0.42409	67	87	22.99%	
	V12	0.47738	0.47390	0.48098	0.48000	108	115	6.09%	
	All	0.46614	0.45801	0.47467	0.46427	1025	1198	14.44%	
40%	V01	0.16977	0.10001	0.28839	0.10474	30	110	72.73%	55.77%
	V02	0.16722	0.10682	0.26188	0.11438	15	46	67.39%	
	V03	0.19442	0.14486	0.26096	0.14433	42	90	53.33%	
	V04	0.20442	0.17707	0.23600	0.17250	56	105	46.67%	
	V05	0.16615	0.13389	0.20618	0.12626	57	148	61.49%	
	V06	0.19135	0.13771	0.26587	0.14189	51	122	58.20%	
	V07	0.21561	0.17215	0.27009	0.17652	75	154	51.30%	
	V08	0.21683	0.17808	0.26408	0.17686	101	192	47.40%	
	V09	0.18710	0.10897	0.32127	0.11076	4	16	75.00%	
	V10	0.25155	0.22578	0.28028	0.21137	8	13	38.46%	
	V11	0.18189	0.12892	0.25676	0.13379	32	87	63.22%	
	V12	0.19288	0.14647	0.25407	0.15039	57	115	50.43%	
	All	0.19832	0.15263	0.25773	0.15328	528	1198	55.93%	

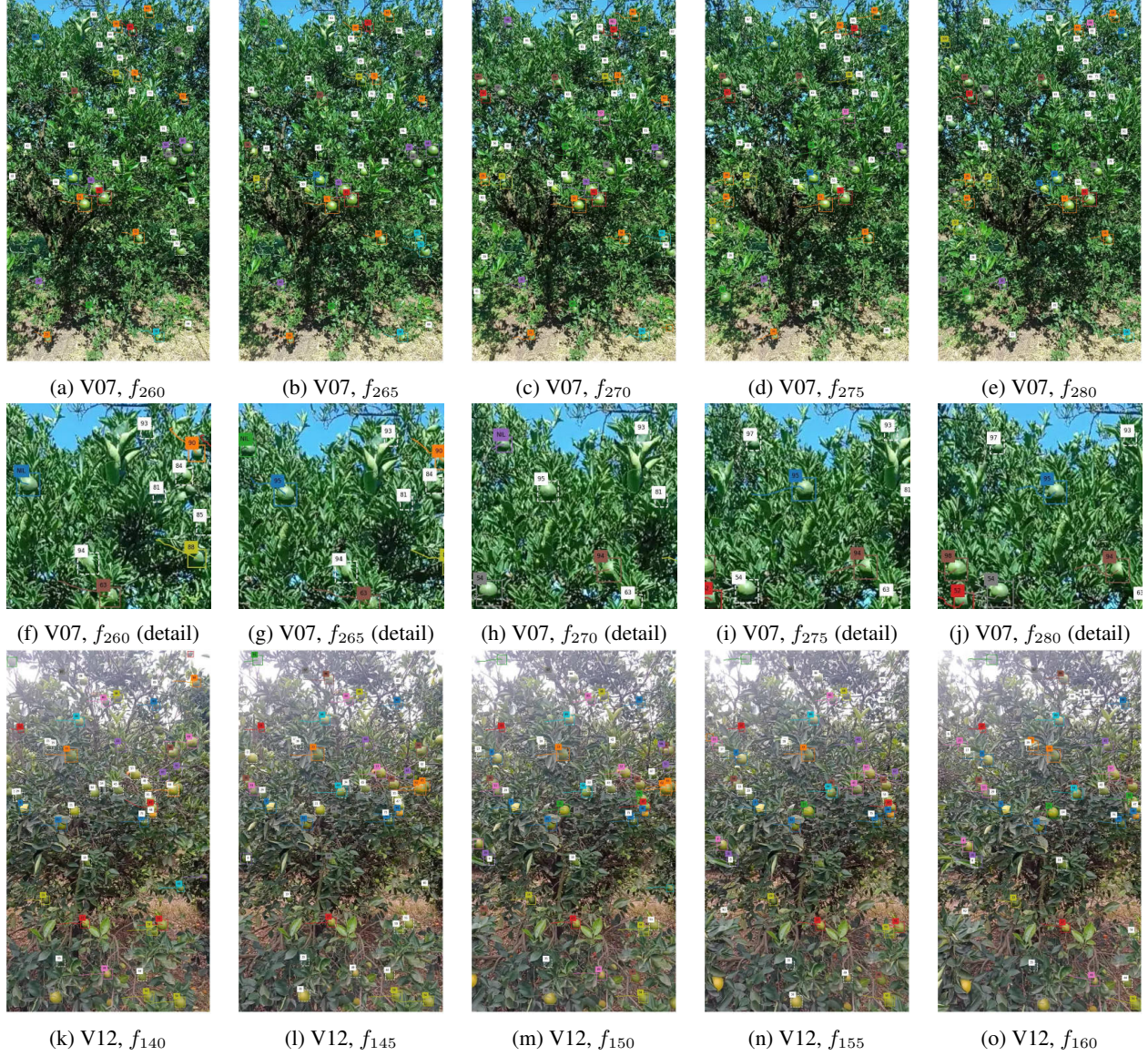


Figure 5: Tracking results for inputs presenting 80% fruit detection ratio. Tracks whose oranges' 3-D models were estimated present numerics IDs. New tracks whose oranges were not yet estimated successfully presents the marker NIL. Tracks marked by dashed white lines are LOST tracks. (a-j) Frames from V07 (*Valencia*). (k-o) Frames from V12 (*Hamlin*). Best seen in digital format.

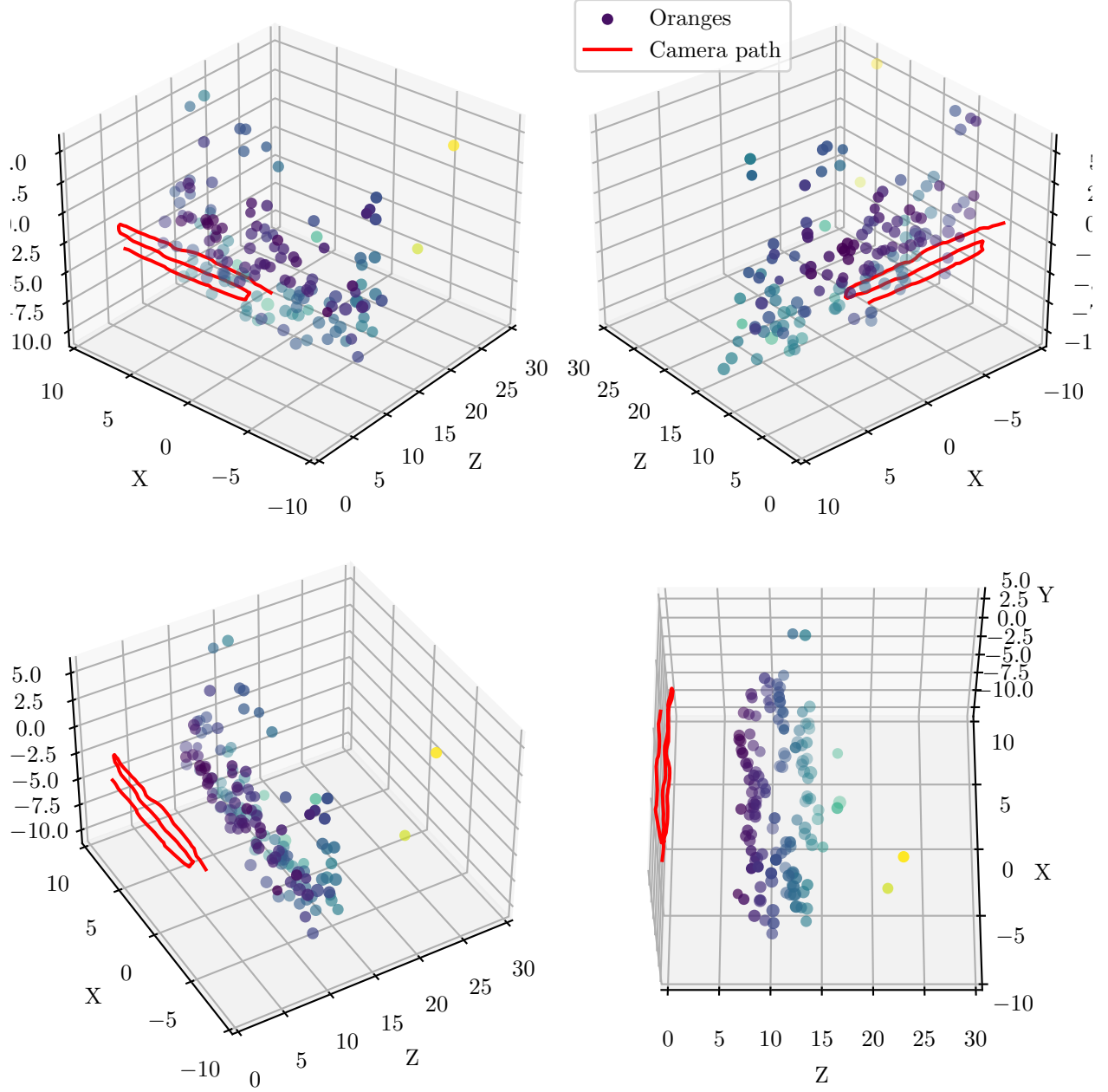


Figure 6: Estimations for oranges' 3-D localizations in V07. MOT was performed using an 80% detection rate. Scale is arbitrary (SfM without metric scale calibration). The camera path is shown in red. The 145 found oranges are shown as circles, the color scale representing the distance in the Z axis (depth). The four plots show different angles of the same structure.

Table 6: Sensitivity analysis without relocation. The table shows how tracking (and fruit counting) degrades with detection performance if the tracker does not incorporate relocation.

Detection (%)	Sequence	HOTA	DetA	AssA	MOTA	CbyT	CbyT-GT	Relative error (%)
100%	V01	0.34960	0.44284	0.27629	0.44890	244	110	121.82%
	V02	0.43208	0.54064	0.34543	0.55949	139	46	202.17%
	V03	0.48072	0.50524	0.45741	0.51485	150	90	66.67%
	V04	0.40903	0.42646	0.39232	0.43890	169	105	60.95%
	V05	0.35899	0.55269	0.23319	0.52738	581	148	292.57%
	V06	0.46115	0.45874	0.46365	0.47716	183	122	50.00%
	V07	0.40752	0.40948	0.40565	0.42184	276	154	79.22%
	V08	0.45201	0.44898	0.45514	0.46420	289	192	50.52%
	V09	0.42806	0.31271	0.58611	0.31669	14	16	12.50%
	V10	0.40208	0.42290	0.38235	0.43222	20	13	53.85%
	V11	0.48725	0.44734	0.53124	0.46841	101	87	16.09%
	V12	0.48120	0.50184	0.46146	0.51958	171	115	48.70%
	All	0.43426	0.46100	0.40914	0.47157	2337	1198	95.08%
80%	V01	0.17219	0.24220	0.12258	0.23293	272	110	147.27%
	V02	0.16378	0.29666	0.09050	0.28548	176	46	282.61%
	V03	0.13913	0.24712	0.07838	0.22648	328	90	264.44%
	V04	0.13699	0.25693	0.07307	0.23968	436	105	315.24%
	V05	0.17829	0.33577	0.09468	0.29965	575	148	288.51%
	V06	0.15788	0.24045	0.10378	0.23092	368	122	201.64%
	V07	0.13617	0.25015	0.07417	0.23541	609	154	295.45%
	V08	0.13391	0.24435	0.07344	0.22780	766	192	298.96%
	V09	0.12673	0.15943	0.10075	0.14665	37	16	131.25%
	V10	0.14627	0.22291	0.09606	0.21137	36	13	176.92%
	V11	0.16579	0.23478	0.11728	0.22643	214	87	145.98%
	V12	0.15895	0.25511	0.09913	0.24286	359	115	212.17%
	All	0.15013	0.25560	0.08826	0.23942	4176	1198	248.58%

4.2 Detection, tracking and counting

Table 7 shows the detection results for seven different orange models. The YOLOv5, YOLOv6, YOLOv7, and YOLOv8 networks can use different backbone sizes. We have tested the *small* and *large* versions of these networks, the latter producing the best results, so only the *large* (l) models are shown. Two different backbone sizes are also tested for EfficientDet: the B0 (baseline) and B3 backbones [Tan et al., 2020]. The results in Table 7 consider a minimum score of 0.5 for the orange class and an IoU of 0.5 between predicted boxes and ground-truth. The values are *not* an average over images in the test set: they were computed considering the *entire* set of 121,685 bounding boxes (oranges) in the ORANDET test set, being an assessment of all fruit misses and false positives.

As shown in Table 7, YOLOv5l, YOLOv8l and YOLOv7l presented close results considering the F_1 -score, followed by YOLOv3. However, there is a slight difference in their behavior: the YOLOv3 and YOLOv8l networks present better precision, a lower number of false positives, while the YOLOv5l and YOLOv7l models show better recall, missing fewer fruits. Note that the sensitivity test presented in Section 4.1 does not consider any level of false positives in the detections, neither any misalignment between boxes in the ground-truth and in the predictions. The different false positive rates of the detection models and their localization accuracy can affect the tracking.

Table 7: Detection results considering a ground-truth composed by 121,685 bounding boxes in ORANDET test set. The displayed results correspond to a score threshold of 0.5 (orange class) and a minimum IoU of 0.5 between detection and ground-truth. TP, FP, and FN correspond to true positives, false positives and false negatives, respectively.

	Detections	TP (%)	FP (%)	FN (%)	Precision	Recall	F_1 -score
YOLOv5l	109,919	69.27	21.06	30.73	0.77	0.69	0.73
YOLOv8l	96,245	64.72	14.37	35.28	0.82	0.65	0.72
YOLOv7l	101,914	66.14	17.61	33.86	0.79	0.66	0.72
YOLOv3	80,092	57.68	8.14	42.32	0.88	0.58	0.70
YOLOv6l	85,997	57.50	13.17	42.50	0.81	0.58	0.67
EfficientDet B3	79,129	55.42	9.61	44.58	0.85	0.55	0.67
EfficientDet B0	70,760	50.66	7.49	49.34	0.87	0.51	0.64

Table 8 shows tracking results for different orange detection models. For each frame, tiling with overlap is employed: the frame is split into 24 tiles with 416×416 pixels, presenting overlaps of 82 pixels. The tiles are stacked in a single

batch, and submitted to the fruit detection CNN. The detection results are merged (“untilling”) and non-maximum suppression (NMS) applied (we have adopted 0.2 as IoU threshold for NMS). We have also filtered detections by score, testing three different score thresholds (0.5, 0.6 and 0.7). Table 8 displays the five best tracking results, considering the score threshold that produced the best result for such detection model. The results for each frame sequence in MORANGET for the two best models (YOLOv5l and YOLOv3) are shown in Table 9. It is noteworthy that when the tracking results are integrated, i.e., when the 1,198 fruits/tracks are considered, the counting error rates are low.

Table 8: Tracking results, considering the ground-truth in MORANGET. The relative error is computed for the entire set of tracks in the 12 frame sequences, summing 1,198 tracks. The value in the last column (Median) is computed by evaluating the relative error for each video and getting the median (see Table 9).

	HOTA	DetA	AssA	MOTA	CbyT	Relative error (%)	Median
YOLOv5l (score > 0.7)	0.56039	0.48968	0.64591	0.58540	1,183	1.25%	6.95%
YOLOv3 (score > 0.5)	0.51003	0.43885	0.59608	0.54786	1,153	3.76%	8.45%
YOLOv8l (score > 0.6)	0.54216	0.46512	0.63690	0.56415	1,128	5.84%	9.40%
YOLOv7l (score > 0.6)	0.52306	0.45016	0.61086	0.53131	1,140	4.84%	10.24%
EfficientDet B3 (score > 0.5)	0.55564	0.47085	0.66561	0.56775	1,086	9.34%	11.82%

Figure 7 shows tracking examples for two videos, V07 and V12, the same previously seen in Figure 5 for comparison. The tracking was performed using detections from the YOLOv5l model. Again, when the orange position and ray were estimated by Algorithm 2, the numeric ID for the track (fruit) is displayed, and tracks not yet presenting a successful estimation are marked with a NIL label. LOST tracks are displayed as white, dashed boxes. A relocalization example can be seen for track 111 in V07: the track is lost at frames f_{260} and f_{265} , but the track became ACTIVE at f_{270} . A compilation of the results in a video demonstration is available on-line¹³.

Perfect tracking implies exact counting and a HOTA value equal to 1. As seen in Table 5, high HOTA (or MOTA) values are associated with low errors in counting. However, we can observe low errors in counting even for mediocre values of HOTA (or MOTA), around 0.5, as seen in Table 8. Actually, multiple fruit tracking is a *harder* problem than fruit counting assessment. If the pipeline can track the fruit in *part* of its trajectory and does not merge tracks of different fruits, it can reach accurate counting values. But losing parts of the fruit’s trajectories will severely penalize the tracking quality measures. If the tracker is losing part of the trajectories, it is missing the orange appearances in some frames. Figure 8 shows histograms for four frame sequences in MORANGET, considering the number of appearances for each fruit in the ground-truth (orange colored histogram) and in the tracker results (blue colored histogram). More “mass” in bins on the right side means the fruits are observed non-occluded in more frames. Comparing the ground-truth against the tracking results (using YOLOv5l detection in this example), we can see that our pipeline missed parts of the tracks, even considering accurate counting: the relative counting errors are 0.91%, 3.81%, 3.65% and 0.87% for V01, V04, V08 and V12, respectively, but the missed appearances will degrade HOTA and MOTA values. Our pipeline can keep fruits in the LOST state, passive of relocalization, so the oranges, even missed in some frames, are properly accounted. Figure 9 shows an example where motion blur jeopardized detection: the fruits are kept in the LOST state, but HOTA and MOTA values decrease because ground-truth consider them visible oranges (note as part fruits are properly relocalized in the next frame).

4.3 Yield regressor results

The sample analyzed consisted of the 1,197 plants that resulted from the previous steps of the pipeline. However, some plants had to be excluded due to missing plant dimensions (W , H or D in Table 3) or lack of automatic fruit counting, resulting in a reduced set of 1,139 plants. The regressor was specifically designed to estimate the sum of fruits from the first to the third flowering, namely $F1 + F2 + F3$, as the fruits from the fourth flowering are too small to be detected in the previous stages of the pipeline.

Consider the counting from our tracking system fed by YOLOv3 orange detections. Out of the 1,139 fruits, 911 were used for training of the neural network regressor, and the remaining 228 for testing were reserved for testing. The results are presented in Figure 10, where we can observe a significant dispersion of points in the graph. This dispersion affected the overall performance, resulting in a R^2 value of 0.61. Notably, there is an isolated point in the lower part of the graph where the regressor estimated around 50 fruits, while the ground truth falls within the range of 1,250 to 1,500. Conversely, there are two isolated points at the top of the graph where the regressor overestimated the values. For these points, the regressor estimated values between 1,500 and 1,750, while the actual values are around 700 for the left point and around 1,200 for the right point.

¹³<https://youtu.be/h0q42KMskLQ>

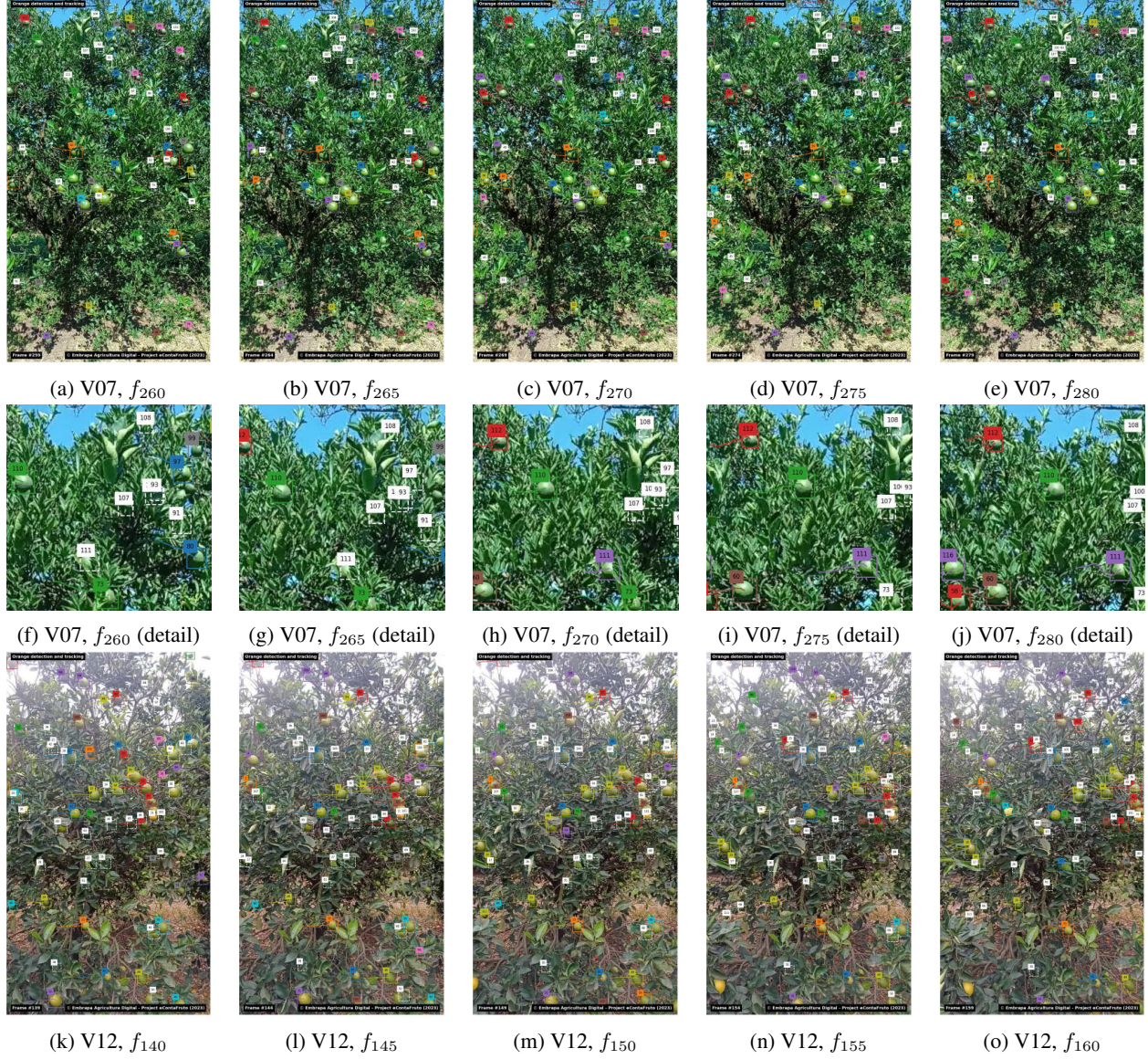


Figure 7: Tracking results for detection inputs from YOLOv5l. Tracks whose oranges' 3-D models were estimated present numerics IDs. New tracks whose oranges were not yet estimated successfully presents the marker NIL. Tracks marked by dashed white lines are LOST tracks. (a-j) Frames from V07 (*Valencia*). (k-o) Frames from V12 (*Hamlin*). Best seen in digital format.

Table 9: Tracking results per frame sequence in MORANGET for the two best detection models: YOLOv5l and YOLOv3.

Model	Sequence	HOTA	DetA	AssA	MOTA	CbyT	CbyT-GT	Relative error (%)	Median
YOLOv5l	V01	0.37662	0.32271	0.47037	0.06078	111	110	0.91%	6.95%
	V02	0.34375	0.24189	0.48938	0.17777	34	46	26.09%	
	V03	0.43321	0.31249	0.60213	0.34024	81	90	10.00%	
	V04	0.71576	0.69602	0.73719	0.84919	109	105	3.81%	
	V05	0.60224	0.60302	0.60454	0.72446	151	148	2.03%	
	V06	0.51853	0.45734	0.59217	0.55306	146	122	19.67%	
	V07	0.66271	0.63850	0.68908	0.78735	148	154	3.90%	
	V08	0.55260	0.48412	0.63734	0.62637	185	192	3.65%	
	V09	0.36263	0.22681	0.58042	0.28029	10	16	37.50%	
	V10	0.23916	0.12979	0.44391	-0.02332	15	13	15.38%	
	V11	0.52879	0.45788	0.61570	0.58926	77	87	11.49%	
	V12	0.53879	0.46801	0.62259	0.59311	116	115	0.87%	
All		0.56039	0.48968	0.64591	0.58540	1183	1198	1.25%	
YOLOv3	V01	0.33805	0.27482	0.42379	0.20005	95	110	13.64%	8.45%
	V02	0.36436	0.25440	0.52319	0.27699	33	46	28.26%	
	V03	0.41799	0.32480	0.53953	0.34516	94	90	4.44%	
	V04	0.59680	0.55379	0.64500	0.71497	111	105	5.71%	
	V05	0.58442	0.56636	0.60592	0.73024	152	148	2.70%	
	V06	0.52529	0.46520	0.59506	0.56571	139	122	13.93%	
	V07	0.53014	0.48528	0.58142	0.64985	136	154	11.69%	
	V08	0.50786	0.43300	0.60138	0.56000	175	192	8.85%	
	V09	0.31171	0.16878	0.57590	0.18565	7	16	56.25%	
	V10	0.30117	0.18895	0.48078	0.05758	14	13	7.69%	
	V11	0.50065	0.41959	0.60027	0.52388	80	87	8.05%	
	V12	0.54416	0.47418	0.62598	0.60044	117	115	1.74%	
All		0.51003	0.43885	0.59609	0.54786	1153	1198	3.76%	

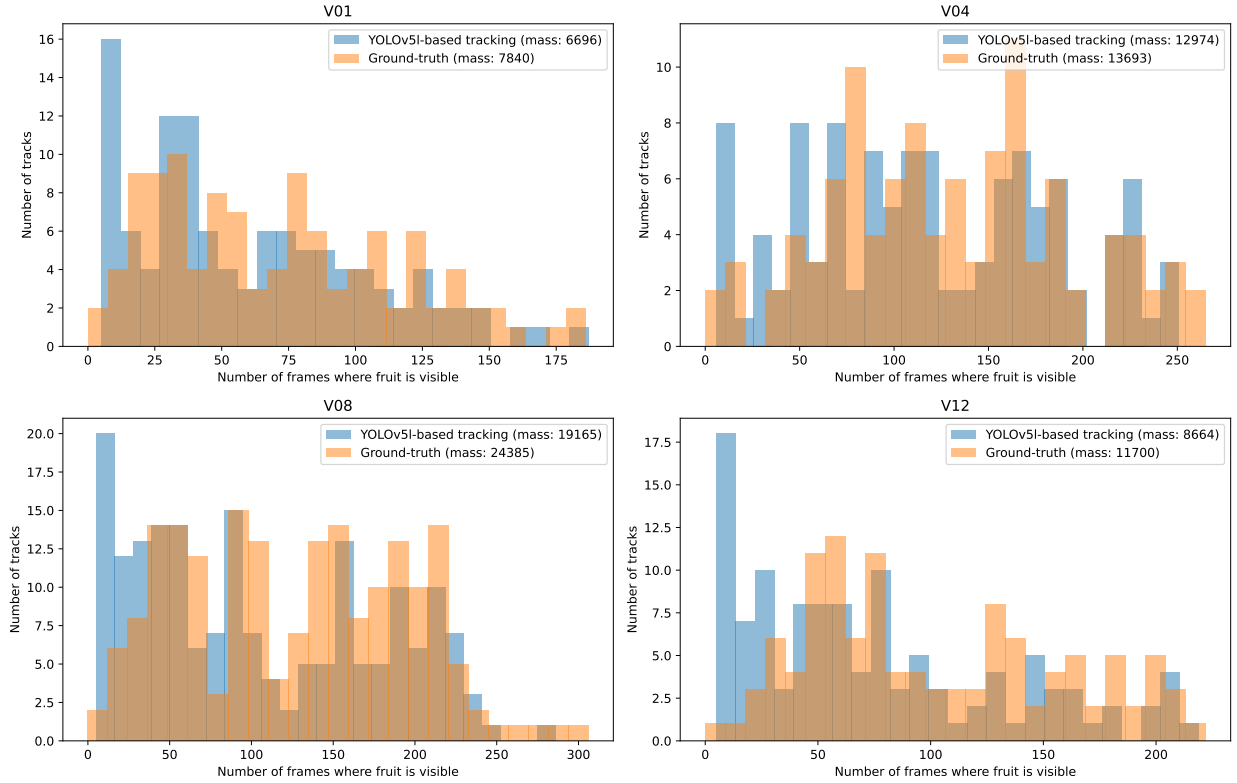


Figure 8: Histograms showing the accumulated number of oranges appearances, contrasting the ground-truth (orange histograms) and the pipeline tracking results (blue histograms) for four frame sequences in MORANGET. More mass in the histograms means more appearances of fruits.

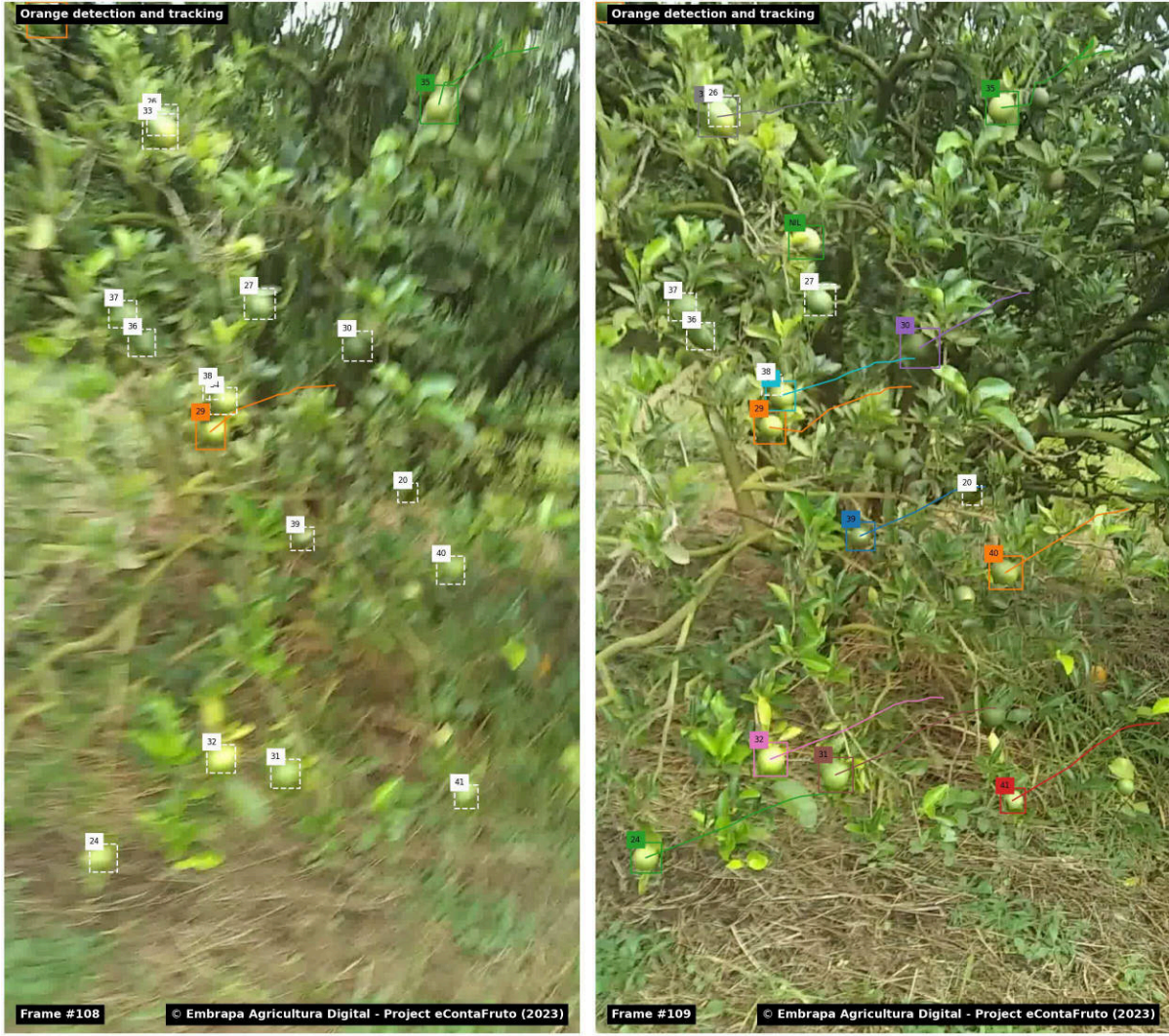


Figure 9: Fruits in LOST state during tracking (dashed white boxes) in frame 108 from V01. The misdetections are caused by the motion blur in the frame. Our tracking module follows the tracks, keeping them in LOST (non-visible) state, while ground-truth data considers such tracks visible fruits. Part of the tracks are relocalized in the next frame 109. Best seen in digital format.

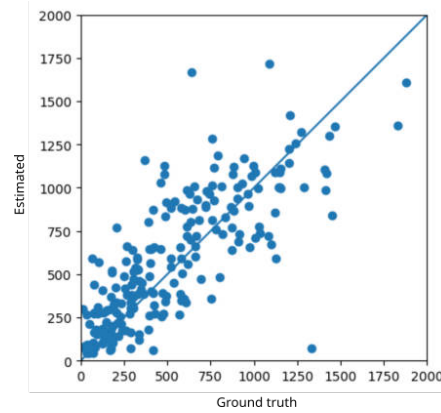


Figure 10: Processing results considering the entire dataset, achieving performance of $R^2 = 0.61$.

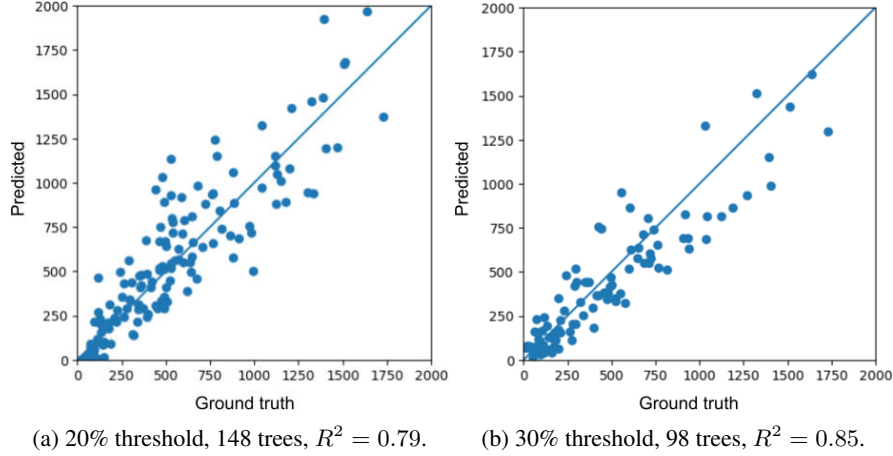


Figure 11: Yield regression, subject to different thresholds for fruit counting. (a) Ensuring at least 20% of the ground truth identified by counting (148 trees in the test set), $R^2 = 0.79$. (b) Ensuring at least 30% of the ground truth identified by counting (98 trees in the test set), $R^2 = 0.85$. The counting was performed by tracking from YOLOv3.

During our analysis of the full dataset, which includes the fruit counting from videos captured on each side of the plant (CbyT-A + CbyT-B), the ground truth ($F1 + F2 + F3$), and the yield estimated by the regressor, we observed that a significant number of samples had a much lower number of counted visible fruits (CbyT-A + CbyT-B) compared to the sum $F1 + F2 + F3$. In fact, most of these samples lie below the 40% detection rate mentioned in Table 5.

This analysis led us to develop the hypothesis that implementing an acceptance threshold for the videos, based on the ratio between detected versus ground truth fruits, could potentially improve the results of the regressor. To test this hypothesis, we established thresholds for the acquisition process. Initially, we considered only videos with a minimum identification rate of 20% of the ground truth fruits. This new experiment reduced the dataset to 741 plants, with 148 plants reserved for testing purposes.

The results of this experiment showed a significant improvement, with the R^2 value increasing to 0.79, as illustrated in Figure 11a. We can also observe a reduced dispersion around the diagonal line, indicating better performance. This improved value of the R^2 suggests that our neural network regressor was able to explain a greater portion of the variance in yield estimation when considering this reduced dataset.

Furthermore, when we raised the threshold to require the detection of at least 30% of the ground truth (still below the 40% in Table 5), the R^2 value further improved to 0.85, as shown in Figure 11b. This new experiment reduced the dataset to 558 trees, with 98 plants reserved for testing purposes, seen in Figure 11b. These results clearly indicate the performance of the regressor is closely tied to the quality of the machine learning identification and fruit counting results. It is important to acknowledge that it is not possible to identify 100% of the fruits through an automatic counting process, as some fruits are located inside the canopies, not visible by camera in any pose.

We observe the same behavior when considering the counting produced using tracking from YOLOv5l detections. In this case, requiring detection of at least 30% of the ground truth, we got a set of 668 trees. Of these, 557 were chosen for training the regressor, resulting in a R^2 value of 0.82 for the test set, which comprised the remaining 111 trees. A scatter plot of this result is seen in Figure 12.

5 Discussion

The complex nature of the natural environment, coupled with the variability in fruit farming practices, continues to pose significant challenges for automating both monitoring and harvesting. As noted by He et al. [2022a], achieving automated operations in complex agricultural fields requires a synergistic approach that combines engineering and standardization of agricultural management practices. Such integration is crucial to enable future automated operations and management. It is essential that breeders and agronomists adapt the plants for automated operation. For example, the results by Gao et al. [2022] show the clear advantages of systems like *fruiting wall* on automated counting and yield estimation. Our pipeline reached significant results in a diverse set of orchard management, using hand-held video streams recorded by smartphones. We expect even better results in more controlled settings, like fruiting walls and artificial lightning at night operations, and employing improved hardware like wide-angle lenses, and visual-inertial

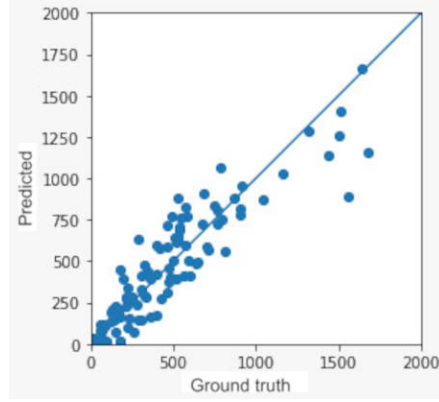


Figure 12: Yield regression for a test set of 111 trees, subject to a threshold of at least 30% of the ground truth identified by counting. The counting was performed by tracking from YOLOv5l detections. The yield regression reached $R^2 = 0.82$.

cameras. The image capture procedure employed in this work is coupled to the distributed nature of the Fundecitrus’ Orange Crop Forecast, performed by 30 field agents covering a large territory. Video recordings by smartphones were a practical, cheap solution, after other designs involving UAVs and cameras arrays being considered, and prior tests performed on field.

The Crop Forecast employs a well-established methodology [Fundecitrus, 2022], relying on a large set of individual trees rather than the scanning of entire rows or plots. One challenge arising from this tree-based approach is the difficulty in visually defining the boundaries of neighboring trees, leading to branches from one tree entering the canopy of another. While the Crop Forecast team can manage this issue through manual fruit stripping, enabling accurate attribution of oranges to the correct tree (albeit a laborious task for the forecast staff), a vision-based system would find error-free fruit-to-tree assignment impractical. Other approaches, such as those integrating row-level or plot-level data and utilizing vehicle-mounted cameras, as demonstrated by Zhang et al. [2022], could be considered. In row-level counting, exact fruit-to-tree attribution would not pose an issue.

Besides the large number of works on fruit detection and impressive results reached by neural networks-based systems, fruit detection is still challenging. Considering large datasets containing a representative set of images, and presenting diversity in light conditions, sensors, noise, crop varieties and phenological stage, detection performance can yet be significantly improved. However, accurate detection is necessary but not sufficient for accurate fruit counting, mainly because of occlusion issues. Better detection models depend on large annotated datasets for training and evaluation, but the scarcity of such datasets remains a key bottleneck in developing the next-generation of intelligent systems for precision agriculture, as pointed by Lu and Young [2020]. We hope that MORANGET and ORANDET be significant contributions in this regard.

Kalman filter-based trackers [Bewley et al., 2016] can be considered an alternative to handle short-term fruit lost, but it is not as efficient as relocalization for long occlusion periods and changes in the movement direction. Our relocalization module can be considered an alternative to the motion displacement estimation proposed by Zhang et al. [2022], specially useful for non-linear movements and non-planar fruit spatial distributions. Zhang et al. [2022] argues that SORT-like tracking [Bewley et al., 2016, Wojke et al., 2018] cannot handle dense fruit spatial distribution in space and heavy overlap. We have shown that relocalization based on 3-D is a viable alternative to complement inter-frame fruit tracking by data association algorithms. Full 3-D reconstruction is not needed, just ego-motion, which can be obtained by other methods like visual-inertial odometry, benefiting of new odometry-capable camera hardware that recently became commercially available.

As seen in Table 9, some sequences present large relative errors. Consider sequences V09 and V10, that show two faces of the same small orange tree under a challenging setting: wind is shaking the branches, occlusion by leaves is severe, and oranges are in a more mature stage than the examples in the training set. Considering less than 20 oranges are visible, a few missing oranges in these examples represents a large relative error. In this challenge setting, both the short-term tracking component, inter-frame association, and the long-term tracking component, 3-D relocalization, are jeopardized. Noteworthy that when the tracking results are integrated, i.e., when the 1,198 fruits/tracks are considered, the counting error rates are low. This is specially important considering that a yield prediction system must integrate several trees for accuracy. Despite large errors regarding single trees, the results point to tracker stability when considering a larger number of oranges.

Multiple fruit tracking is a harder problem than fruit counting. Seeking for fruits in images, accurately identifying them, and evaluating when they are visible/reachable and when not is a challenge. Our reported values for HOTA and MOTA, and the results reported by de Jong et al. [2022] and Villacrés et al. [2023] indicate MOT in orchards research still needs significant improvements to achieve high tracking rates, above 0.9. However, accurate fruit counting could be achieved before highly accurate MOT. It is noteworthy that fruit tracking on the field is a challenging task even for humans.

Our yield regression confronts a significant challenge: understanding the relationship between observable and unobservable fruits. Koirala et al. [2021] delved into a similar aspect concerning mango trees. Initially, our yield regression displayed a modest prediction capability, with $R^2 = 0.61$. However, ensuring that a minimum of 30% of the fruits were accurately identified by the computer vision-based counting system elevated the R^2 values beyond 0.80. Yet, the validity of assuming that at least 30% of the yield is visible in the canopy, amenable to vision-based counting, warrants scrutiny. Is this assumption realistically grounded? We anticipate that sweet orange trees predominantly bear fruit in the outer or upper regions of the canopy due to enhanced sunlight exposure.

Upon scrutinizing the MORANGET dataset, compiled from high-quality video recordings, it became evident that in most sequences the oranges identified by annotators corresponded to 30 to 50% of the true yield ($F1 + F2 + F3$) obtained through fruit stripping. An exception was noted in sequence V03, where only 12.5% of the fruits were discerned by annotators. This serves as evidence that low-quality video input significantly contributed to errors, and accurate regression from visible fruits holds promise. Moreover, this underscores the need for advancements in image acquisition systems, potentially incorporating solutions like artificial lighting during night operations and employing high dynamic range image sensors.

6 Conclusion

This work proposes a complete pipeline for yield estimation in citrus trees, comprising imaging methodology for fruit detection, multiple orange tracking and yield regression integrating fruit counting to other tree data as size and age. Exploiting ego-motion data, i.e., camera pose information, we were able to create (i) a practical process for ground-truth annotation, and (ii) a relocalization module for multiple fruit tracking, able to deal with long-term occlusions and fruit entering and exiting the camera’s field of view. Framed as a multiple object tracking (MOT) problem, fruit counting was developed and evaluated utilizing established MOT metrics such as MOTA and HOTA. A useful intermediary result is the 3-D fruit localization that could be employed in other tasks and analysis, as robotic harvesting. Orange trees under standard crop management practices will present non-visible fruit, hidden in the deeper parts of the canopy. A yield regressor, that takes to account tree size and age besides crop variety, in addition to image-based fruit counting data, was developed and validated considering a large set of trees (> 1,000) from a real crop forecast effort in one of the largest sweet orange production sites in the world. In this challenging scenario, our pipeline was able to reach an R^2 up to 0.85 to true yield from fruit stripping. Even higher values for R^2 could be reached for better quality video input. Future prospects for this research involve adapting the pipeline for deployment in vehicles and exploring contemporary deep learning-based models tailored for object tracking in image sequences.

CRedit authorship contribution statement

Thiago T. Santos: Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization. **Kleber X. S. de Souza:** Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization. **João Camargo Neto:** Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Review. **Luciano V. Koenigkan:** Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Review. **Alécio S. Moreira:** Data Curation, Writing – Review and Editing. **Sônia Ternes:** Project administration, Methodology, Investigation, Validation, Data Curation, Writing – Review and Editing.

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A HOTA

Consider an assignment matrix $\hat{\mathbf{A}}_i^{(\alpha)}[p, q]$ that matches bounding boxes $\mathbf{b}_i^{(p)} \in \mathcal{B}_i$ to *ground-truth boxes* $\hat{\mathbf{b}}_i^{(q)} \in \hat{\mathcal{B}}_i$, the set of annotated bounding boxes for frame f_i , ensuring that $\text{IoU}(\mathbf{b}_i^{(p)}, \hat{\mathbf{b}}_i^{(q)}) \geq \alpha$. The assignments are one-to-one: a box $\mathbf{b}_i^{(p)}$ can be assigned to at most one box $\hat{\mathbf{b}}_i^{(q)}$ in the ground-truth and vice versa. As seen in Section 3.3, $\hat{\mathbf{A}}_i^{(\alpha)}[p, q] = 1$ in the case of assignment and zero otherwise. Let $\mathbf{m}_i^{(p, q)} = \langle \mathbf{b}_i^{(p)}, \hat{\mathbf{b}}_i^{(q)} \rangle$ be a compact representation for an assignment. The set of *true positives* $\text{TP}_i^{(\alpha)}$ for frame f_i is defined as:

$$\text{TP}_i^{(\alpha)} = \{\mathbf{m}_i^{(p, q)} = \langle \mathbf{b}_i^{(p)}, \hat{\mathbf{b}}_i^{(q)} \rangle \mid \hat{\mathbf{A}}_i^{(\alpha)}[p, q] = 1\} \quad (9)$$

Integrating the true positives for all M frames, we have the set $\text{TP}^{(\alpha)}$:

$$\text{TP}^{(\alpha)} = \bigcup_{i=1}^M \text{TP}_i^{(\alpha)} \quad (10)$$

Similarly, the *false negatives* set $\text{FN}_i^{(\alpha)}$ for f_i and the set of all false negatives $\text{FN}^{(\alpha)}$ are defined as:

$$\text{FN}_i^{(\alpha)} = \{\hat{\mathbf{b}}_i^{(q)} \mid \sum_p \hat{\mathbf{A}}_i^{(\alpha)}[p, q] = 0\} \quad (11)$$

(bounding boxes in the ground-truth not matched to any detected box) and

$$\text{FN}^{(\alpha)} = \bigcup_{i=1}^M \text{FN}_i^{(\alpha)} \quad (12)$$

Finally, the *false positive* set $\text{FP}_i^{(\alpha)}$ for f_i and the set of all false positives $\text{FP}^{(\alpha)}$ are defined as:

$$\text{FP}_i^{(\alpha)} = \{\mathbf{b}_i^{(p)} \mid \sum_q \hat{\mathbf{A}}_i^{(\alpha)}[p, q] = 0\} \quad (13)$$

(detected boxes not assigned to any box in the ground-truth) and

$$\text{FP}^{(\alpha)} = \bigcup_{i=1}^M \text{FP}_i^{(\alpha)} \quad (14)$$

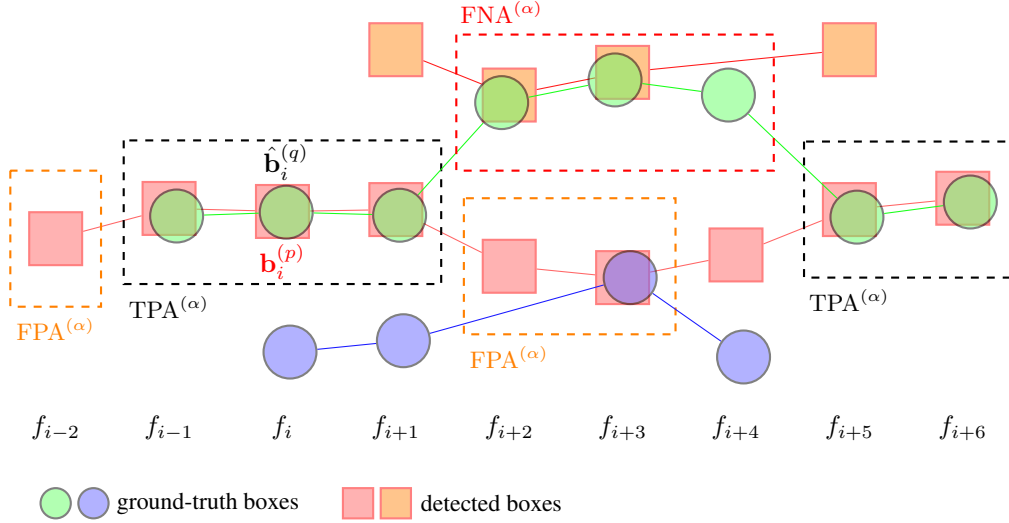


Figure 13: The three association sets in HOTA: true positive association ($\text{TPA}^{(\alpha)}$), composed by pairs (matches) of detected and ground truth boxes (marked in dashed black lines); false negative association ($\text{FNA}^{(\alpha)}$) composed by false negatives and matches with a different detection ID (red dashed lines), and false positive association ($\text{FPA}^{(\alpha)}$), composed by false positives and matches with different ground-truth ID (orange dashed lines). Ground-truth boxes are displayed as circles for visualization purposes. The colors represent the different IDs for ground-truth and detected boxes. Based on Figure 2 in Luiten et al. [2021].

Assume $\text{gid}(\cdot) : \bigcup_i \hat{\mathcal{B}}_i \rightarrow \mathbb{N}$ a function that maps boxes in the ground-truth to numeric identifiers, and $\text{pid}(\cdot) : \bigcup_i \mathcal{B}_i \rightarrow \mathbb{N}$ a function that does the same for detected boxes. To evaluate tracking in HOTA, Luiten et al. [2021] proposed three novel concepts. For a matching $\mathbf{m}_i^{(p,q)}$, the *true positive associations* set $\text{TPA}^{(\alpha)}$ is composed of all true positives $\mathbf{m}_j^{(r,s)}$ that present the same IDs for prediction and ground-truth as $\mathbf{m}_i^{(p,q)}$:

$$\text{TPA}^{(\alpha)}(\mathbf{m}_i^{(p,q)}) = \{\mathbf{m}_j^{(r,s)} \in \text{TP}^{(\alpha)} \mid \text{pid}(\mathbf{b}_i^{(p)}) = \text{pid}(\mathbf{b}_j^{(r)}) \wedge \text{gid}(\hat{\mathbf{b}}_i^{(q)}) = \text{gid}(\hat{\mathbf{b}}_j^{(s)})\} \quad (15)$$

The *false negative associations* set $\text{FNA}^{(\alpha)}$ is composed of matches that present the same ground-truth identifier $\text{gid}(\hat{\mathbf{b}}_i^{(q)})$ that $\mathbf{m}_i^{(p,q)}$, but detections attributed to different IDs, plus the set of false negatives also identified by $\text{gid}(\hat{\mathbf{b}}_i^{(q)})$:

$$\begin{aligned} \text{FNA}^{(\alpha)}(\mathbf{m}_i^{(p,q)}) = & \{\mathbf{m}_j^{(r,s)} \in \text{TP}^{(\alpha)} \mid \text{pid}(\mathbf{b}_i^{(p)}) \neq \text{pid}(\mathbf{b}_j^{(r)}) \wedge \text{gid}(\hat{\mathbf{b}}_i^{(q)}) = \text{gid}(\hat{\mathbf{b}}_j^{(s)})\} \\ & \cup \{\hat{\mathbf{b}}_j^{(s)} \in \text{FN}^{(\alpha)} \mid \text{gid}(\hat{\mathbf{b}}_i^{(q)}) = \text{gid}(\hat{\mathbf{b}}_j^{(s)})\} \end{aligned} \quad (16)$$

The last concept is the *false positive associations* set $\text{FPA}^{(\alpha)}$, composed of matches that present the same detected identifier $\text{pid}(\mathbf{b}_i^{(p)})$, but whose ground-truth presents different identifiers, plus all false positives also identified by $\text{pid}(\mathbf{b}_i^{(p)})$:

$$\begin{aligned} \text{FPA}^{(\alpha)}(\mathbf{m}_i^{(p,q)}) = & \{\mathbf{m}_j^{(r,s)} \in \text{TP}^{(\alpha)} \mid \text{pid}(\mathbf{b}_i^{(p)}) = \text{pid}(\mathbf{b}_j^{(r)}) \wedge \text{gid}(\hat{\mathbf{b}}_i^{(q)}) \neq \text{gid}(\hat{\mathbf{b}}_j^{(s)})\} \\ & \cup \{\mathbf{b}_j^{(r)} \in \text{FP}^{(\alpha)} \mid \text{pid}(\mathbf{b}_i^{(p)}) = \text{pid}(\mathbf{b}_j^{(r)})\} \end{aligned} \quad (17)$$

Figure 13 illustrates the three kinds of association sets. Finally, HOTA_α can be defined as

$$\text{HOTA}_\alpha = \sqrt{\frac{\sum_{\mathbf{m}_i^{(p,q)} \in \text{TP}^{(\alpha)}} \mathcal{A}(\mathbf{m}_i^{(p,q)})}{|\text{TP}^{(\alpha)}| + |\text{FN}^{(\alpha)}| + |\text{FP}^{(\alpha)}|}} \quad (18)$$

where $\mathcal{A}(\mathbf{m}_i^{(p,q)})$ measures the alignment between predicted and ground-truth tracks:

$$\mathcal{A}(\mathbf{m}_i^{(p,q)}) = \frac{|\text{TPA}^{(\alpha)}(\mathbf{m}_i^{(p,q)})|}{|\text{TPA}^{(\alpha)}(\mathbf{m}_i^{(p,q)})| + |\text{FNA}^{(\alpha)}(\mathbf{m}_i^{(p,q)})| + |\text{FPA}^{(\alpha)}(\mathbf{m}_i^{(p,q)})|} \quad (19)$$

Luiten et al. call this a *double Jaccard* formulation: the Jaccard metric is employed on the evaluation of detection, with the matches $\mathbf{m}_i^{(p,q)} \in \text{TP}^{(\alpha)}$ being weighted by the association score $\mathcal{A}(\mathbf{m}_i^{(p,q)})$, which also is a Jaccard metric. The proponents also argue that the metric is the geometric mean of a detection score and an association score, considering that

$$\text{DetA}_\alpha = \frac{|\text{TP}^{(\alpha)}|}{|\text{TP}^{(\alpha)}| + |\text{FN}^{(\alpha)}| + |\text{FP}^{(\alpha)}|} \quad (20)$$

$$\text{AssA}_\alpha = \frac{1}{|\text{TP}^{(\alpha)}|} \sum_{\mathbf{m}_i^{(p,q)} \in \text{TP}^{(\alpha)}} \mathcal{A}(\mathbf{m}_i^{(p,q)}) \quad (21)$$

$$\begin{aligned} \text{HOTA}_\alpha &= \sqrt{\frac{\sum_{\mathbf{m}_i^{(p,q)} \in \text{TP}^{(\alpha)}} \mathcal{A}(\mathbf{m}_i^{(p,q)})}{|\text{TP}^{(\alpha)}| + |\text{FN}^{(\alpha)}| + |\text{FP}^{(\alpha)}|}} \\ &= \sqrt{\text{DetA}_\alpha \cdot \text{AssA}_\alpha} \end{aligned} \quad (22)$$

HOTA_α evaluates detection and association, but not the localization accuracy for the matches, i.e., the spatial fit between boxes' areas. Localization is evaluated by the final HOTA score, that integrates HOTA_α for different values of α :

$$\text{HOTA} = \int_0^1 \text{HOTA}_\alpha d\alpha \approx \frac{1}{19} \sum_{\alpha \in [0.05, 0.1, \dots, 0.9, 0.95]} \text{HOTA}_\alpha \quad (23)$$

Equations 15 to 23 correspond to Luiten et al. [2021] formulations, rewritten to employ the notations adopted in the present work. It is important to note that, like in MOTA [Bernardin and Stiefelwagen, 2008], the employed assignments $\hat{\mathbf{A}}_i^{(\alpha)}[p, q]$ are the ones that maximize the final HOTA score (see Luiten et al. [2021] for details about the matching optimization procedure). Here, we will employ HOTA for the evaluation of our multiple orange tracking, and DetA and AssA scores to decompose tracking assessment into its detection and association components (note that DetA and AssA metrics are computed from DetA_α and AssA_α integrating different α thresholds, as seen for HOTA in Equation 23). We have employed the original code from HOTA authors¹⁴.

¹⁴<https://github.com/JonathonLuiten/TrackEval.git>.